The Market Reads the News First: A Pre-Registered, Look-Ahead-Free Test of AI Decision Models on Stock-Price Reactions*†

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October 2026

Abstract

Can AI models that read financial news predict how stock prices will react, or tell when the market has over- or under-reacted? We test this with AI decision models, non-generative language models that return probabilities for typed questions, in a pre-registered, look-ahead-free design on S&P 500 stocks. We fine-tune an open-weights decision model on 2024–2025 news and returns, compare it with commercial decision models and standard baselines, mask company identities, freeze the analysis plan with a cryptographic hash, and unblind a January–July 2026 test period once. The fine-tuned model's signal replicated on two validation periods (daily information coefficient 0.065, $t = 3.6$) but fell to 0.020 ($t = 1.7$) on the blinded test; none of the five primary hypotheses is supported. Where the models predict anything, it is the overnight gap: commercial models forecast the move to the open (information coefficient 0.074 to 0.077) but not the move after it. The models cannot tell over- from under-reaction, and no pre-registered trading strategy survives transaction costs and a deflated Sharpe ratio. Commercial models identify masked companies in 62% to 74% of cases against a 28% baseline, a warning about look-ahead bias in backtests of large language models (LLMs) on historical news. In an exploratory analysis, conditions selected on two validation periods, with in-sample Sharpe ratios up to 6.8, failed on the test. The market reads the news first.

Keywords: large language models (LLMs); AI decision models; stock return prediction; financial news; look-ahead bias; memorization; data contamination; backtest overfitting; deflated Sharpe ratio; pre-registration; overnight returns; market efficiency; overreaction and underreaction; calibration.

JEL codes: G14, G12, C45, C52, C12.

*Intended arXiv categories: cs.AI (primary); cross-lists q-fin.ST, cs.LG and cs.CL.

†Data and code availability: **[data and code statement and the URL of the replication package, to be provided by the authors]**.

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1 Introduction

Much company news is published before the stock market opens. Two questions follow. Can a model that reads that news anticipate how the price will react? And once part of the reaction is visible, can it tell whether the market moved too far or not far enough? Both are old questions in empirical finance. Prices react to firm news within minutes (Patell and Wolfson, 1984; Busse and Green, 2002), media tone predicts short-horizon returns (Tetlock, 2007), and moves that come with news tend to drift while moves without news tend to reverse (Chan, 2003). Large language models (LLMs) now read news at scale, and several studies find that their readings predict returns (Lopez-Lira and Tang, 2026; Chen et al., 2022). We ask whether such predictability survives a test built to rule out the usual ways a backtest misleads: look-ahead in the data, memorization in the model and selection over many variants.

We study *AI decision models*. A decision model is a non-generative model that takes a *state* (text plus structured context) and a set of typed questions (a choice among options, a score on an ordered scale, or a yes-or-no question) and returns probabilities over the answers instead of generating text (Zhu et al., 2026). Decision models are close relatives of LLM-based classifiers and of probabilistic forecasting with language models (Kadavath et al., 2022; Halawi et al., 2024; Schoenegger et al., 2024), and their vendors describe the probabilities as calibrated. They suit finance because they return the quantities a forecaster needs, such as a distribution over the size of the coming move or the probability that a piece of news is material, at a cost of a few cents per thousand decisions, and because open-weights versions can be fine-tuned on free cloud GPUs.

Language models also bring a specific threat. A model trained on text published after the period on which it is tested may know how events turned out, and even a model with an earlier training cutoff can recognize a company from the content of its news (Glasserman and Lin, 2024; Sarkar and Vafa, 2024; Lopez-Lira et al., 2025). A backtest of a language-model signal on historical news therefore mixes the model's ability to read news with its memory of what followed. A second threat is older: when many signals, filters and horizons are tried, the best backtest overstates what a strategy will earn (Harvey et al., 2016; Bailey and López de Prado, 2014). Pre-registration, fixing the analysis before the data are seen, is the standard remedy in other sciences and is still rare in empirical finance (Harvey, 2017; Olken, 2015).

We combine these elements in a pre-registered, look-ahead-free test. The sample is the point-in-time S&P 500 from July 2024 to July 2026. For each company and trading session we build the news state as of 09:15 ET and mask company names, tickers, people, dates and every mention of a price move. Before the open, each model gives a *pre-open reading*: how the stock will perform once the news is absorbed. At 10:00 and after the close we show the model the reaction so far and ask for a *post-reaction reading*: was the reaction too small or too large for the news? The primary instrument, Laya-Fin, is an open-weights decision model that we fine-tuned on 2024–2025 news and returns. We compare it with three commercial decision models chosen in a zero-shot tournament, the untuned Laya model, an anchor model trained in the same way from an encoder released before the sample, and standard baselines. Every definition, hypothesis and selection rule was written down before any outcome was computed; selection used two validation quarters of 2025; the analysis plan was frozen with a SHA-256 hash and a version-control tag before any 2026 model call; and the 2026 outcomes were unblinded once, through a lock that logs every read. Every evaluated variant is logged as a trial, and every deviation from the plan is logged with its date and reason. The study cost $15.73 in API fees and 25.4 GPU-hours on a free cloud tier.

Key findings.

(1) **Clean validation did not survive the blinded test.** The fine-tuned model's daily information coefficient (IC) with the close-to-close move was 0.055 and 0.065 ($t = 3.6$) in the two 2025 validation quarters but 0.020 ($t = 1.7$) in the 2026 test; none of its five primary hypotheses is supported, and the commercial models' ICs roughly halved as well (d1-ultra from 0.097 to 0.039).

(2) **Pre-open news is priced at the open.** The commercial decision models' daily IC with the overnight move from the previous close to the open is 0.074 to 0.077 ($t = 4.5$ to 5.3), and their IC with the move after the open is about zero (-0.001 to 0.002); no model predicts the move after the open.

(3) **The models cannot tell over- from under-reaction.** In the pre-registered over/under regression the news coefficient is not positive (Laya-Fin $b = -0.044$, $p = 0.62$) and the reversal coefficient has the predicted sign without significance ($c = -0.015$, $p = 0.11$). Prices reach part of the model's news-implied value by the close (a slope of 0.36 for Laya-Fin and 0.59 for d1-ultra), then partly fade, and the gap's half-life is not reached within 20 trading days.

(4) **Commercial decision models recognize "anonymized" firms.** Given masked news, they pick the company out of four same-sector candidates in 62% to 74% of cases, against a matched baseline of 28%; the open-weights Laya model does so in 32%. This is a warning for backtests of language models on historical news.

(5) **News explains little of a stock's own move out of sample.** Same-day news explains 0.25% of the stock-specific daily variance for Laya-Fin (supported; Benjamini–Hochberg $q = 0.058$) and 0.82% for d1-ultra. Reading the realized move explains 15.8%, but the moves the model calls explained do not persist. The models know which stocks will move a lot (an AUC of 0.60), but no better than the stocks' own recent volatility (an out-of-sample R^2 gain of 0.0012 over a realized-volatility benchmark).

(6) **The commercial models are overconfident.** Laya-Fin is calibrated but compressed (expected calibration error 0.057, log-loss skill 0.001), while the commercial models' stated probabilities overstate their hit rates (expected calibration error 0.209 to 0.361).

(7) **No pre-registered strategy survives costs and selection.** No strategy clears its deflated Sharpe ratio (the largest probability is 0.024, with 545 logged trials) or the 95th percentile of its random-signal placebo, and the convergence strategy is effectively a price-only gap fade (92% of positions shared, daily-return correlation 0.96).

Contributions. First, we give a protocol for testing language-model signals on financial news without look-ahead: point-in-time states, masking checked by an independent scanner, a pre-registration frozen by hash before any test-period call, a lock that releases test outcomes once, and ledgers of spend, trials and test-data access, all in a replication package. Second, we show where the predictable part of pre-open news goes: into the opening price. Third, we measure how prices converge to a model's news-implied value and read the convergence path through an errors-in-variables lens that separates what its level and its shape identify. Fourth, we add two measurement tools for language models in finance: a matched-distractor test of whether a model recognizes masked companies, and a firm-level version of the explanation ladder of Kelly et al. (2026), which measures how much of a stock's own daily move its news explains.

Section 2 reviews related work and Section 3 describes the data and the design. Sections 4 to 9 report the pre-registered results, nulls included. Section 10 reports exploratory analyses, labelled post hoc, and Section 11 the limitations and the pre-registered forward test.

2 Related work

Language models and stock returns. Text has long been used to predict returns, from dictionary measures of the tone of news (Tetlock, 2007; Tetlock et al., 2008) to sentiment scores learned from returns (Ke et al., 2019). Large language models extend this line: Lopez-Lira and Tang (2026) find that ChatGPT's reading of headlines predicts next-day returns, and Chen et al. (2022) find that LLM representations of news predict the cross-section of returns. Kelly et al. (2026) use generative AI to explain realized market returns with the reasons given in daily market summaries, revisiting Roll (1988), who found that market, industry and public news explain only a small part of individual stock returns; we carry their explanation ladder to the firm level. Decision models are a newer interface (Zhu et al., 2026): the model returns probabilities for typed questions instead of text, so no free text needs parsing and every output is a forecast that can be scored.

Look-ahead bias and memorization. Glasserman and Lin (2024) test for look-ahead bias in GPT's sentiment readings of news by anonymizing the companies; Sarkar and Vafa (2024) analyze look-ahead bias in pretrained language models used for prediction; Lopez-Lira et al. (2025) show that LLMs recall economic and financial data from their training period, so forecasting tests inside that period are not out of sample; and He et al. (2026) train language models only on text available at each date. In machine learning, memorization and benchmark contamination are documented and measured directly (Carlini et al., 2023; Golchin and Surdeanu, 2024; Sainz et al., 2023). Every decision model we use was released after our sample ended, so none can be assumed ignorant of it. We therefore mask the states, measure how often a model still identifies the company, and pre-register a forward test on news published after the models' release.

News, attention and price reactions. Prices incorporate firm news quickly (Patell and Wolfson, 1984; Busse and Green, 2002) but not always completely. Earnings news drifts (Bernard and Thomas, 1989); price moves that come with news continue while moves without identifiable news reverse (Chan, 2003; Savor, 2012); high-frequency evidence points to underreaction to news (Jiang et al., 2021); and identifying the relevant news sharpens the link between information and volatility (Boudoukh et al., 2019). Stale or recombined news is mispriced (Tetlock, 2011; Fedyk and Hodson, 2023), and distracted or inattentive investors underreact (Hirshleifer et al., 2009; DellaVigna and Pollet, 2009). Behavioral models explain both underreaction and overreaction (Barberis et al., 1998; Daniel et al., 1998), building on the overreaction evidence of De Bondt and Thaler (1985), while Fama (1998) argues that the two kinds of anomaly are about equally frequent and fragile to the choice of method. The split of a daily return into an overnight and an intraday part (Lou et al., 2019) is central here: we ask on which side of the open the news is priced.

Selective prediction and calibration. A model that can abstain trades coverage for accuracy (Geifman and El-Yaniv, 2017), and whether its confidence is a useful guide depends on calibration, which modern neural networks often lack (Pakdaman Naeini et al., 2015; Guo et al., 2017). Language models are partly calibrated about what they know (Kadavath et al., 2022), and LLM forecasts can approach the accuracy of human forecasters (Halawi et al., 2024; Schoenegger et al.,

2024). We measure calibration with the pre-registered expected calibration error and use selective prediction only in the exploratory section.

Pre-registration and replication in finance. Many published return predictors do not survive multiple-testing adjustments or replication (Harvey et al., 2016; Hou et al., 2020), although Jensen et al. (2023) find that most factors replicate once the evidence is aggregated properly. Harvey (2017) calls for pre-registration and other safeguards in financial economics, and Olken (2015) discusses the benefits and limits of pre-analysis plans in economics. For trading strategies, the deflated Sharpe ratio adjusts a backtest for the number of trials behind it (Bailey and López de Prado, 2014). We pre-register hypotheses, families and corrections, log every trial, and report every pre-registered result.

Our earlier work. Kargarzadeh et al. (2026) ask when news is priced in, using millions of news articles on about three thousand US stocks, each tagged by an LLM-distilled classifier with an event type and several attributes. They find that the reaction is concentrated before and on the day of publication, that markets underreact to numbers and overreact to stories, and that news carries width as well as direction. This paper extends that work in three ways. It replaces the event tags with decision models that read the masked news and forecast the move directly; it splits the reaction at the open and at fixed decision points, so that it can ask whether a reader of the news can trade on it; and it moves from a descriptive study to a pre-registered, blinded test with a single unblinding.

3 Data and design

The pre-registration, its amendments and its deviations log are part of the replication package. Appendix A summarizes the log (Table A1), and Appendix B collects the notation (Table A2).

3.1 Universe, sample periods and returns

The universe is the point-in-time membership of the S&P 500: a stock is in the sample on a session only if it was an index member on that session, and companies with several share classes count once. Each month the members are split into three size tiers by market capitalization at the previous month-end: the 100 largest, ranks 101–250 and ranks 251–500. The sample is split by time into a training period (2024-07-01 to 2025-06-30), two validation quarters and a test period. The first validation quarter, val-A (2025-07-01 to 2025-09-30), is used for search and for fitting calibration maps, temperatures and thresholds; the second, val-B (2025-10-01 to 2025-12-31), for confirmation and the final selection. The test period, 2026-01-02 to 2026-07-31, was blinded until the analysis plan was frozen and was then used once. A target enters an evaluation only if its whole horizon ends inside its period, and the first five sessions of each evaluation period are left out.

Daily prices are adjusted for splits, dividends and spin-offs, and the opening price is put on the same basis. For stock i on session t, the close-to-close return splits at the open into an overnight return, from the previous close to the open, and an intraday return, from the open to the close:

$$1 + r^{cc}_{i,t} = (1 + r^{on}_{i,t})(1 + r^{id}_{i,t}), \qquad z^{k}_{i,t} = \frac{\mathrm{AR}^{k}_{i,t}}{\sigma^{k}_{i,t-1}}, \quad k \in \{cc, on, id\}. \tag{1}$$

In logs, the close-to-close move is the sum of the overnight move and the move after the open. Each part is made abnormal by removing the market and the stock's sector, $\mathrm{AR}^{k}_{i,t} = r^{k}_{i,t} - \beta^{m}_{i}\, r^{k}_{m,t} -$

$\beta_i^s\,(r_{s,t}^k - r_{m,t}^k)$, where r_m is the return of the S&P 500 ETF (SPY), r_s the equal-weighted return of the other members of the stock's sector, and the betas come from the 252 sessions up to $t-1$. It is then divided by $\sigma_{i,t-1}^k$, a robust scale (1.4826 times the median absolute deviation) of the stock's abnormal returns over the 126 sessions up to $t-1$. A value of $z_{i,t}^k$ is therefore a move in units of the stock's typical daily move, scaled by information known before the session. Equation (1) defines what "priced at the open" means in this paper: news is priced at the open when a model's reading predicts z^{on} but not z^{id}. Where the indices are clear we write z_{cc}, z_{on} and z_{id}, as the tables and figures do. Multi-day targets add up abnormal returns over their window and divide by the window's ex-ante volatility; the main one, y_2, runs from the next open to the close of $t+5$.

3.2 News, events and decision points

The news comes from a commercial company-news feed (titles and snippets with publication times). An article is available at its time stamp, or at the end of the hour or day when the stamp is rounded to the hour or carries only a date. We drop items marked as non-news, price commentary and promotion, automatic institutional-holdings reports, non-English articles, near-duplicates, and two kinds of item whose time stamp precedes its content: explainers of a stock's move and running market stories updated in place. We fix three decision points per session t:

- **D0**, at 09:15 ET, before the open. The model reads the news published since the previous close; a trade enters at the official open.
- **D1**, at 10:00 ET, after the first 30-minute bar. The model also sees the move so far; a trade enters at the price at the end of that bar.
- **D2**, after the close. The model reads the day's news and the day's move; a trade enters at the next open.

The observed reaction G is the move up to the decision point in typical-move units: at D2 it is the day's z_{cc}, and at D1 the abnormal move from the previous close to 10:00, divided by its ex-ante volatility. Its level version G^{lvl} divides by σ^{cc} without a square-root-of-time adjustment, so that moves at different horizons add up.

A learned importance filter, a logistic regression on sentence embeddings of the news (Xiao et al., 2024), selects the company-days that enter the analysis. A pre-registered rule chose it on val-A among seven candidate filters: at a budget of 30% of covered company-days it captures 43.5% of the company-days with a material price reaction. An *event* is a company-session with selected news: D0 events use the news up to 09:15, D2 events the news up to the close. The test period has 13,748 D0 events on 140 sessions and 17,188 D2 events.

Each event becomes a *state*, a JSON document with fixed keys: the company's context (sector, size tier, market sensitivity, volatility regime, recent abnormal performance, analyst-estimate revisions and whether an earnings report is scheduled), the market's context (the previous session's market move, the level and change of the VIX, rates, credit spreads, the dollar, oil and scheduled macroeconomic releases), attention (the number of articles against the stock's usual number) and up to six news items with their source type and timing. Every number passes through a codebook of words: a move of between one and two typical daily moves is "moderate", followed by its multiple of a typical day's move. A post-reaction state adds the reaction so far: the stock's move, the market's move over the same period and the volume against normal.

The two readings. The *pre-open reading* asks, among other questions, "Once investors fully absorb the news, how will the company's stock perform relative to the overall market and its sector

over the next five trading days?" on nine levels, from a very large fall (more than three typical daily moves) to a very large rise. With p_k the probability of level k and m_k the level's midpoint in typical daily moves, the *news-implied move* is the model's expected move,

$$\hat{N} = \sum_{k=1}^{9} p_k m_k, \qquad m = (-3.5,\ -2.5,\ -1.5,\ -0.75,\ 0,\ 0.75,\ 1.5,\ 2.5,\ 3.5).$$

The *news-implied value* V maps $\hat{N}$ to the realized five-day level move with an isotonic regression (Barlow et al., 1972) fitted on val-A for each model: it is the model's estimate of where the abnormal price settles once the news is absorbed. The *post-reaction reading* also asks whether the reaction so far is in the wrong direction, much too small, too small, about right, too large or much too large for what the news justifies, and whether the move will reverse, stall or continue over the next five days. Its judgment $J = P(\text{much too small}) + P(\text{too small}) - P(\text{too large}) - P(\text{much too large})$ is positive when the model calls the reaction an underreaction. A pre-registered check found that the post-reaction readings anchor on the move the model is shown, so $\hat{N}$ always comes from the pre-open reading.

Masking. The states hide the company. Names, aliases, tickers and cashtags become "the company"; person names become "an executive" or "a person"; dates, weekdays, years and fiscal-period labels become "recently" or "the latest quarter"; share-price levels and every percent price move of any security (the company's, other companies', indexes, commodities and currencies) are removed. Fundamental figures, such as earnings per share, revenue and guidance, are kept, because they are the news. An independent leak scanner, written without the masker's rules, and reading of the remaining candidates checked the result; the residues that remain were accepted below half a percent of states for names, tickers and dates, and are disclosed in the deviations log. Masking cannot hide a company whose news is distinctive, and Section 8 measures how often the models still recognize it.

3.3 The models

Laya-Fin, the primary instrument. Laya is an open-weights decision model built on the ModernBERT-large encoder (Convai Innovations, 2026; Warner et al., 2025). We fine-tuned it on the training period's states with three heads: the nine-level move (labelled with the realized five-day move), a five-level magnitude head (the size of the same-day move) and, for the post-reaction reading, a three-class continuation head. Training ran for one epoch on two T4 GPUs of Kaggle's free tier, with AdamW (Loshchilov and Hutter, 2019) and early stopping on val-A, and one temperature per head was fitted on val-A. The pre-registered plan called for three seeds; one seed (13) was trained before the freeze, and the other seeds are pending robustness runs (Section 11).

Commercial decision models. Every eligible decision model in a public model router's catalogue, plus the local Laya models, entered a two-stage zero-shot tournament on val-A: 21 models answered the same 700 states. A model confidence set (Hansen et al., 2011) kept all of them, because about a dozen events a session cannot separate daily ICs. The six with the highest first-stage IC then answered 3,000 states and faced four pre-registered screens: company identification on masked states (pass at 40% or less, against 25% by chance), the share of counterfactual edits (beat or miss, raise or cut, upgrade or downgrade) that move the reading the right way (pass at 70% or more), the IC gain from unmasking, and a shuffled-news control. No model passed: all six identified the masked company in 63% to 72% of states, and five also failed the counterfactual screen. By a

decision logged before the freeze, the three with the highest second-stage IC, d1-ultra, span-02 and drex-v2.0, were carried to the test as *screen-failing, model-specific comparators*: we report their results, but they cannot support a general claim. All three were released in October 2026, after the sample ended.

The untuned model, the anchor and the baselines. Laya zero-shot is the untuned Laya model answering the same questions. The anchor is the December 2024 ModernBERT-large checkpoint (Warner et al., 2025), fine-tuned with the same heads and recipe as Laya-Fin; it was meant to show what training adds to a model whose pretraining ended before the sample. After one epoch its outputs are nearly constant, so we report it as *uninformative*, not as a null, and make no claim about the effect of training. The baselines are vendor sentiment and a gradient-boosted model (Ke et al., 2017) on the vendor's topic tags, both built on labels the vendor produced after the fact and therefore an upper bound that may carry look-ahead, not a benchmark; a ridge regression (Hoerl and Kennard, 1970) on sentence embeddings of the news; a gradient-boosted model on price, volume and size alone; and a generative LLM prompted for the same nine-level forecast on a 2,000-event subsample.

3.4 Pre-registration, freeze and blinding

The pre-registration was written on 2026-10-24, before any panel, target or model output existed. It fixes the universe, the splits, the targets, the masking, the questions, the selection rules for filters, encodings and models, the hypotheses with their families and corrections, the strategies and the inference. Three amendments were added the same day, before any data: convergence to the news-implied value, a wider question battery and the firm-level explanation analysis. Changes after that go only into an append-only deviations log, with the date, the reason and whether they came before or after the freeze. Selection used validation data only.

On 2026-10-28, before any 2026 model call, the analysis plan was frozen: a SHA-256 hash over the pre-registration (up to its deviations log), every configuration file and the code of the targets, the reactions and the analysis specification was recorded and tagged in version control. Prices from 2026 were sealed: they could be read only through a lock that checks the frozen hash and logs every read. The lock was opened after the freeze to build the post-reaction states (reactions only), and once, in a single logged sequence, to release the test outcomes. Three append-only, hash-chained ledgers record API spend (capped at \$30; \$15.73 used), every evaluated variant as a trial (545 when the pre-registered strategies were evaluated, 1,620 after the exploratory study) and every access to test-period data. The deviations log holds 66 entries; Appendix A summarizes the material ones.

3.5 Inference

The main statistic is the daily information coefficient,

$$\mathrm{IC}_t(\hat{N}, z) = \rho_S\big(\{\hat{N}_{i,t}\}_{i\in\mathcal{E}_t},\, \{z_{i,t}\}_{i\in\mathcal{E}_t}\big), \qquad \overline{\mathrm{IC}} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{IC}_t(\hat{N}, z), \tag{2}$$

the Spearman rank correlation ρ_S between the readings and the moves across the events $\mathcal{E}_t$ of session t (sessions with at least five events), averaged over the T sessions of a period. Its t-statistic uses Newey–West standard errors with five lags (Newey and West, 1987). A positive IC means that the readings rank the day's moves in the right order on average; the IC is the usual measure of a cross-sectional signal. The hypotheses that news is priced at the open compare the IC with the

overnight move and with the move after the open:

$$\text{H1: } \Delta = \frac{1}{T}\sum_t \Big[\mathrm{IC}_t(\hat{N}, z^{on}) - \mathrm{IC}_t(\hat{N}, z^{id})\Big] > 0, \qquad \text{H1b: } \frac{1}{T}\sum_t \mathrm{IC}_t(\hat{N}, z^{id}) > 0. \tag{3}$$

H1 says that the news is priced mostly at the open, H1b that it is not fully priced there, so that a position taken at the open can still earn. Both are tested one-sided with a stationary block bootstrap over sessions (Politis and Romano, 1994), with blocks of five sessions and 10,000 draws. H1b was added to the pre-registration before the freeze, because the pre-registered strategy that enters at the open needs it.

The five primary hypotheses (H1, H1b, H2, H3 and H4, defined where they are tested) form one family, corrected by Holm's method at 0.05 (Holm, 1979); the secondary hypotheses form families corrected by Benjamini–Hochberg at $q = 0.10$ (Benjamini and Hochberg, 1995). Regressions are pooled OLS with day-by-tier fixed effects and standard errors clustered by date and by firm (Cameron et al., 2011), with Driscoll–Kraay standard errors (Driscoll and Kraay, 1998) as a check. A primary result counts as a general claim only if it holds with the same sign and $p < 0.05$ for the primary instrument and for the anchor; if it holds for the primary instrument alone, it is model-specific. Because the anchor is uninformative and the commercial models failed the screens, no result in this paper can be more than model-specific.

4 Clean validation did not survive the blinded test

Figure 1 compares each model's daily IC with the close-to-close move on val-B, the quarter used for the final selection, with its IC on the test period. Laya-Fin was selected as the primary instrument because its val-B IC, 0.065 ($t = 3.64$), was the highest among the candidates eligible for that role; on val-A, which it was not trained on, its IC had been 0.055 ($t = 4.30$). On the test it fell to 0.020 ($t = 1.67$), below the lower end of its val-B 95% interval (0.030 to 0.100). The commercial models fell as well, though they stayed significant: d1-ultra from 0.097 to 0.039, span-02 from 0.099 to 0.041 and drex-v2.0 from 0.075 to 0.044 ($t = 3.1$ to 3.7). The untuned Laya model fell from 0.029 to 0.005. The clean benchmarks are near zero in both periods (on the test, -0.003 for the price-only model and 0.004 for the embedding ridge), and the generative LLM's IC on its subsample fell from 0.052 to 0.001. The highest test ICs belong to the two baselines built on vendor labels (0.055 for the tag model and 0.047 for vendor sentiment); because the vendor produced those labels after the fact, we read them as an upper bound that may contain look-ahead, not as a benchmark. Appendix Table A3 lists every model with its cost per thousand decisions.

Table 1 reports the five primary hypotheses. None is supported for Laya-Fin: the Holm-adjusted p-values are 0.269 for H3 and 0.853 for the other four. H1 and H1b, whether the news is priced at the open, are the subject of Section 5 and Table 2, and H2, the over/under regression, of Section 6 and Table 3. H3 asks whether the post-reaction judgment J adds to price-only and news-only features in predicting y_2, the move from the next open to the close of $t+5$; the gain in IC is 0.0038 (one-sided $p = 0.054$). The pre-registered Shapley decomposition of this fit (Shapley, 1953) divides a total that is about zero (the full model's out-of-sample R^2 for y_2 is -0.0017), so its shares carry no information. H4 asks whether the readings of materiality, surprise and expected magnitude predict the size of the day's move, $|z_{cc}|$, beyond a heterogeneous autoregressive model of realized volatility (Corsi, 2009), an earnings-day flag and the size tier; they add 0.0012 to the out-of-sample R^2 ($p = 0.216$). Laya-Fin's magnitude head on its own ranks the size of the move (an AUC of 0.60 for moves of at least one typical day), so the models know which stocks will move a lot, but no better than the stocks' own recent volatility. The anchor's rows are reported beside Laya-Fin's, but its near-constant readings carry no information.

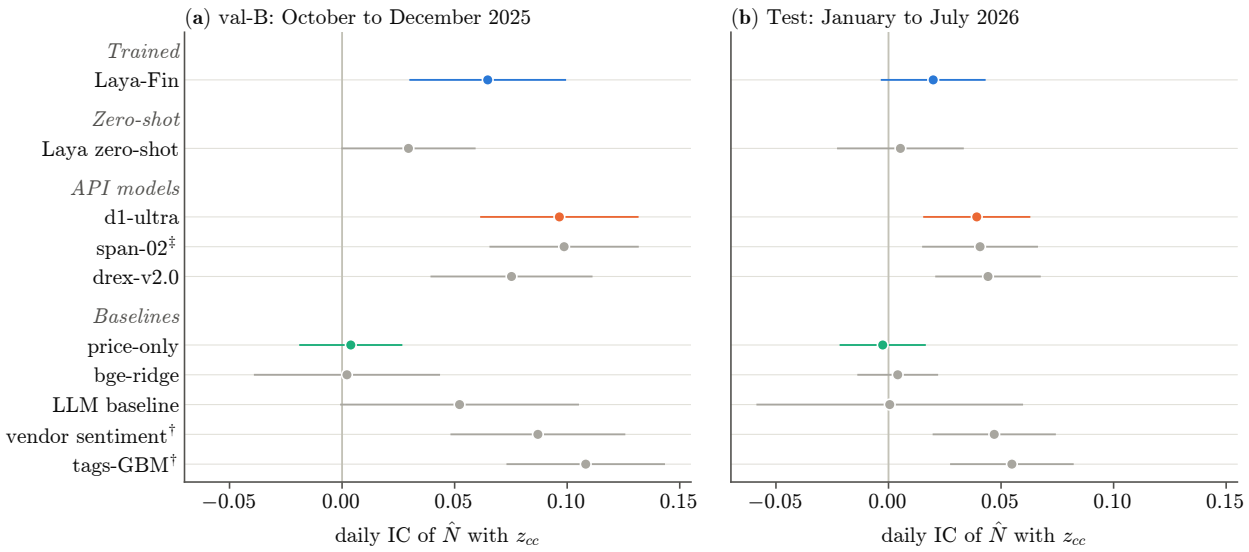


Figure 1: Every validation IC that was positive fell on the blinded test: Laya-Fin's from 0.065 to 0.020, d1-ultra's from 0.097 to 0.039. Daily IC of the pre-open reading $\hat{N}$ with the close-to-close move z_{cc} on (a) val-B (October–December 2025, the final-selection quarter) and (b) the test period (January–July 2026), with 95% intervals (±1.96 Newey–West standard errors); the panels share one scale, with one row per model. The commercial models (API models) failed the pre-registered screens, so their results are model-specific. The clean benchmarks are near zero; the LLM baseline was run on a 2,000-event subsample. †Vendor-label baselines: the labels were produced after the fact, so their ICs are a possibly look-ahead-contaminated upper bound. ‡span-02: a stability check that repeats earlier requests fired after the freeze; its build was unchanged and the results stand (Appendix A). The anchor's near-constant readings are in Table A3.

Table 1: Primary hypotheses on the test period: none of the five is supported for the primary instrument, Laya-Fin (Holm p from 0.269 to 0.853).

	Laya-Fin (primary)				Anchor		
Hypothesis	Estimate	95% CI	p	Holm p	Estimate	p	Verdict
H1	0.005	[−0.035, 0.041]	0.409	0.853	0.004	0.406	not supported
H1b	0.010	[−0.014, 0.037]	0.213	0.853	0.002	0.427	not supported
H2	−0.044	—	0.616	0.853	0.015	0.294	not supported
H3	0.0038	[−0.0008, 0.0085]	0.054	0.269	0.0013	0.017	not supported
H4	0.0012	[−0.0017, 0.0042]	0.216	0.853	−0.0004	0.773	not supported

The tests: H1, $\mathrm{IC}(\hat{N}, z_{on}) - \mathrm{IC}(\hat{N}, z_{id}) > 0$; H1b, $\mathrm{IC}(\hat{N}, z_{id}) > 0$; H2, $b > 0$ and $c < 0$ at D2; H3, ΔIC of J for $y_2 > 0$; H4, ΔR^2 of the news for $|z_{cc}| > 0$.

Test period 2026-01-02 to 2026-07-31, events outside the embargo. Estimates in each hypothesis' direction; p one-sided; 95% CIs and p from a stationary block bootstrap over sessions (block 5, 10,000 draws); Holm over H1, H1b, H2, H3 and H4 at 0.05.

H2: the estimate is $\min(b, -c)$ and $p = \max(p_b, p_c)$ (both must hold), so it has no single interval (—); b and c with their standard errors are in Table 3. Estimates and CIs to three decimals, H3's and H4's to four, since those increments are an order smaller.

Verdict (the pre-registered headline rule): general if Laya-Fin's Holm $p < 0.05$ with the predicted sign and the anchor's own $p < 0.05$ with the same sign; model-specific if only Laya-Fin's holds; otherwise not supported. The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. Its p is its own, not Holm-adjusted.

Significance: no stars are used; p is one-sided unless stated.

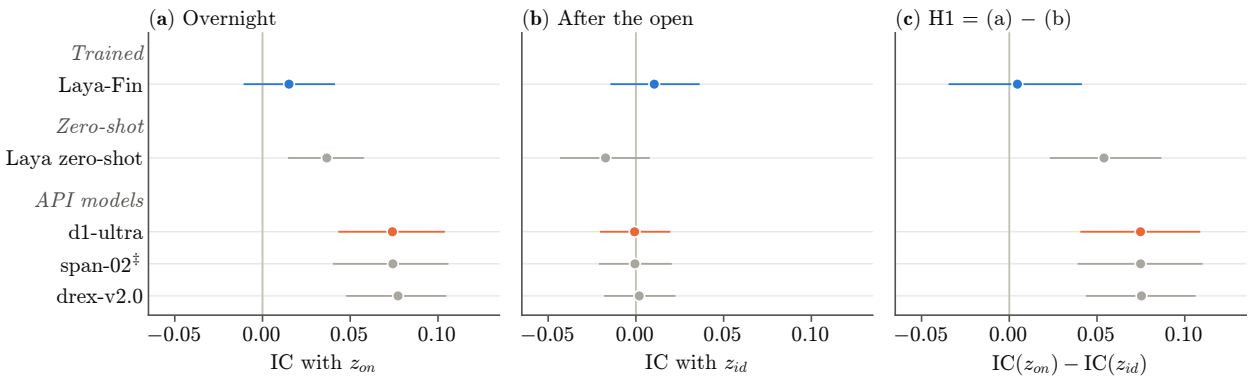


Figure 2: The commercial models' pre-open signal is in the overnight move (daily IC 0.074 to 0.077), not in the move after the open (about zero). Daily IC of the pre-open reading $\hat{N}$ on the test period with (a) the overnight move z_{on} (previous close to open) and (b) the move after the open z_{id} (open to close), with 95% bootstrap intervals; (c) H1, the difference of the two, with its bootstrap interval. The panels share one scale, with one row per model. For d1-ultra, H1 = 0.075 ($p = 0.0001$); the commercial models are model-specific because they failed the pre-registered screens. Laya-Fin's ICs are small on both sides (0.015 and 0.010), so H1 and H1b are not supported for it (Holm $p = 0.853$), although on val-B its IC with z_{id} had been 0.054 ($t = 2.89$). ‡span-02: see Figure 1. The anchor is in Table 1.

The fall from validation to test is not specific to the fine-tuned model: every model whose val-B IC was positive fell, the vendor-label baselines included. An exploratory split in Section 10 shows where the decline happened: away from earnings days. On earnings days, the IC held in all three periods.

5 Pre-open news is priced at the open

Equation (1) splits each day's move at the open. Figure 2 and Table 2 show on which side of the open the models' readings predict it. For the three commercial models the answer is clear. Their daily IC with the overnight move z_{on} is 0.074 (d1-ultra), 0.074 (span-02) and 0.077 (drex-v2.0), with t-statistics of 4.74, 4.49 and 5.34; their IC with the move after the open, z_{id}, is −0.001, −0.001 and 0.002. H1, the difference, is 0.075 for d1-ultra (95% interval 0.040 to 0.109, $p = 0.0001$), and the same to three decimals for the other two. Sorted by d1-ultra's reading, the top quintile of a day's events moves 0.81 typical daily moves more overnight than the bottom quintile ($t = 5.6$). The untuned Laya model shows the same pattern at a smaller scale (H1 = 0.054, $p = 0.001$). Because the commercial models failed the screens, these results are model-specific.

No model predicts the move after the open. H1b, the IC with z_{id}, is not significant for any model on the test: 0.010 for Laya-Fin ($t = 0.8$), and between −0.017 and 0.002 for the others. This matters for Laya-Fin, whose validation signal seemed to reach past the open: on val-B its IC with z_{id} was 0.054 ($t = 2.89$), as large as its IC with z_{on} (0.045). That is why H1b was added to the pre-registration before the freeze. On the test, both of Laya-Fin's ICs are small (0.015 and 0.010) and neither hypothesis is supported.

Robustness. No pre-registered robustness variant changes these conclusions (Appendix Figure A1): dollar-volume tiers, dropping articles with date-only or hour-rounded time stamps, a non-overlapping sample, placebo-adjusted targets, a 60-session volatility, dropping delisting horizons and dropping text about price moves. Two variants speak to when the news arrives. For d1-ultra, the overnight IC is largest when the latest article arrives between 08:00 and 09:15 (0.115,

Table 2: Daily IC by target: the API models' IC is 0.074 to 0.077 with the overnight move and about zero with the move after the open; no model predicts the move after the open.

	Val-B			Test		
Model	z_{cc}	z_{on}	z_{id}	z_{cc}	z_{on}	z_{id}
Trained and zero-shot						
Laya-Fin	0.065	0.045	0.054	0.020	0.015	0.010
	(3.64)	(2.93)	(2.89)	(1.67)	(1.08)	(0.82)
anchor	−0.020	−0.002	−0.011	0.008	0.006	0.002
	(−1.52)	(−0.16)	(−0.75)	(0.83)	(0.50)	(0.18)
Laya zero-shot	0.029	0.047	0.002	0.005	0.037	−0.017
	(1.94)	(2.55)	(0.12)	(0.37)	(3.22)	(−1.25)
API models (model-specific)						
d1-ultra	0.097	—	—	0.039	0.074	−0.001
	(5.38)	—	—	(3.23)	(4.74)	(−0.07)
span-02‡	0.099	—	—	0.041	0.074	−0.001
	(5.83)	—	—	(3.10)	(4.49)	(−0.05)
drex-v2.0	0.075	—	—	0.044	0.077	0.002
	(4.10)	—	—	(3.70)	(5.34)	(0.18)

Mean daily Spearman IC of the pre-open news reading $\hat{N}$ with the close-to-close move z_{cc}, the overnight move z_{on} (previous close to open) and the move after the open z_{id} (open to close), each in typical daily moves, to three decimals; Newey–West t in parentheses, to two; D0 events outside the embargo (val-B 2025-10-01 to 2025-12-31; test 2026-01-02 to 2026-07-31).

Val-B ICs with z_{on} and z_{id} come from a run of the frozen specification on val-B before the freeze, which covered the trained and zero-shot Laya models only (—: not computed).

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the pre-registered contamination screens, so their results are model-specific. ‡span-02: after the freeze, a stability check that repeats earlier requests (the canary) fired for it; its build was unchanged, and production went ahead under a rule recorded in the deviations log.

$t = 5.0$), smaller for 04:00 to 08:00 (0.061) and not significant when the latest article precedes 04:00 (0.024, $t = 1.1$). For Laya-Fin, the IC with z_{cc} is 0.006 when the latest article precedes 04:00 and 0.025 and 0.025 in the two later windows. Unmasking does not raise d1-ultra's IC: on 5,923 test events its IC with z_{cc} is 0.026 with masked states and 0.029 with the same states unmasked. The market reads the news before the open, and for d1-ultra the closer to the open the news arrives, the more of the overnight move its reading predicts.

6 The models cannot tell over- from under-reaction

6.1 The over/under regression

The central pre-registered hypothesis, H2, asks whether a model's reading of the news separates the part of a day's move that will persist from the part that will reverse. After the close (D2) we regress the next week's move on the news-implied move and on the day's move:

$$y_{i,t} = a + b\,\hat{N}_{i,t} + c\,G_{i,t} + \alpha_{d(t)\times\tau(i)} + \varepsilon_{i,t}, \tag{4}$$

where $y_{i,t} = y_2$ is the abnormal move from the next open to the close of $t+5$ in typical-move units, $\hat{N}_{i,t}$ is the model's reading of the day's news (all of it, up to the close, with price language removed), $G_{i,t} = z^{cc}_{i,t}$ is the day's move and $\alpha_{d(t)\times\tau(i)}$ is a day-by-tier fixed effect; there is no conditioning on the size of G. H2 predicts $b > 0$ (moves the news explains persist: underreaction to news) and $c < 0$ (moves the news does not explain reverse: overreaction to price); both must hold, and each is tested one-sided with standard errors clustered by date and firm.

Table 3 reports the estimates. For Laya-Fin, $b = -0.044$ (standard error 0.149, $p = 0.62$) and $c = -0.015$ (0.012, $p = 0.11$) on 17,188 events. The reversal coefficient has the predicted sign for every model but is never significant, and the news coefficient is not positive: d1-ultra's is negative ($b = -0.076$, $p = 0.99$). The two earlier decision points, reported outside the family, do not change the picture: at 10:00, Laya-Fin's b is 0.23 ($p = 0.062$), and before the open, with the move after the open as the outcome, 0.28 ($p = 0.096$). None of the eight pre-registered heterogeneity cuts (three size tiers, quantified against story news, small moves, press wires, low usual coverage and busy days) changes the coefficients for Laya-Fin (Wald p from 0.16 to 0.90, Benjamini–Hochberg $q \geq 0.74$; Appendix Table A4). Nor does the style of the coverage matter. Work on linguistic style treats it as a dimension of text separate from content (Wegmann et al., 2026), and the secondary hypothesis H6 asks whether a more promotional or sensational tone predicts reversal once the content is controlled for; for d1-ultra, the only model asked about tone, the coefficient is 0.010 with the wrong sign (standard error 0.019, $q = 0.95$).

Table 3: The over/under regression (H2): for Laya-Fin the news coefficient b is -0.044 and the reversal coefficient c is -0.015 ($p = 0.110$): c has the predicted sign but is not significant, and b is not positive at the close.

Decision point	b	SE	p_b	c	SE	p_c	Events
Laya-Fin (primary)							
D2, close (H2)	−0.044	(0.149)	0.616	−0.015	(0.012)	0.110	17,188
D1, 10:00	0.229	(0.148)	0.062	0.013	(0.012)	0.861	13,203
D0, open	0.278	(0.212)	0.096	—	—	—	13,439
Anchor (uninformative)							
D2, close (H2)	0.748	(1.377)	0.294	−0.015	(0.012)	0.111	17,188
D1, 10:00	1.964	(2.482)	0.215	0.014	(0.012)	0.874	13,203
D0, open	−4.510	(3.236)	0.917	—	—	—	13,439
d1-ultra (model-specific)							
D2, close (H2)	−0.076	(0.031)	0.991	−0.011	(0.012)	0.171	17,188
D1, 10:00	−0.018	(0.032)	0.711	0.015	(0.012)	0.884	13,203
D0, open	0.022	(0.037)	0.277	—	—	—	13,439

$y = a + b\,\hat{N} + c\,G$ with day × tier fixed effects, on the test period: at D2 $y = y_2$ (next open to t+5), $\hat{N}$ from the day's news and G the day's move; at D1 the move after 10:00 on $\hat{N}$ from the pre-open news and the move to 10:00; at D0 $y = z_{id}$ and there is no G, so no c (—). Two-way clustered (session, firm) standard errors in parentheses; p_b tests $b > 0$ and p_c tests $c < 0$, one-sided. H2 is the D2 row of Laya-Fin.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the pre-registered contamination screens, so their results are model-specific.

Significance: no stars are used; p is one-sided unless stated.

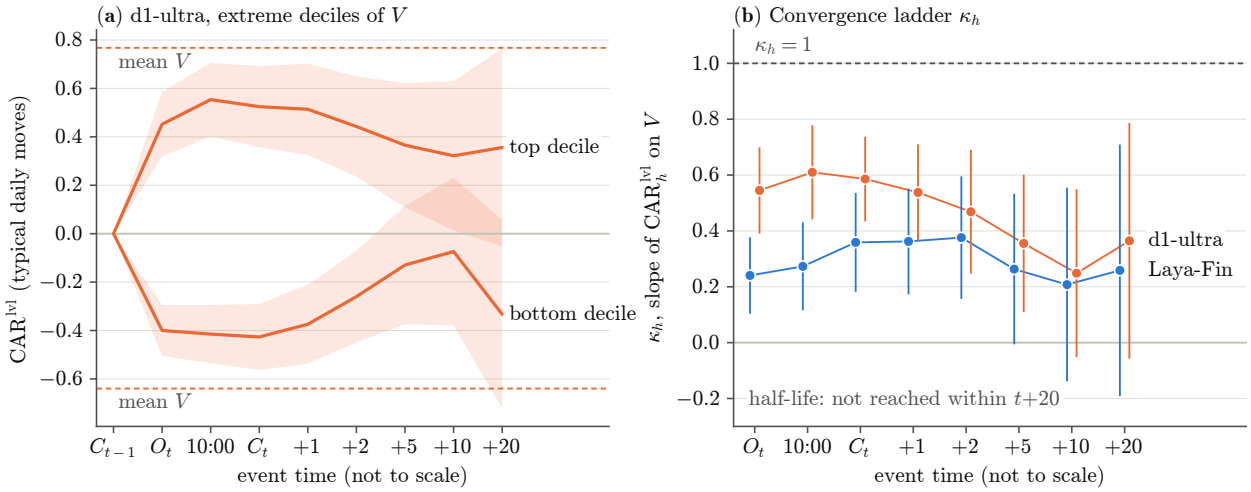


Figure 3: Prices move part of the way to the model's news-implied value by the close, then partly fade: the convergence slope is 0.36 at the close for Laya-Fin and 0.59 for d1-ultra, and the gap's half-life is not reached within 20 sessions. Test period. Event time is not to scale: O_t is the open, 10:00 the D1 decision point, C_t the close of day t and $+k$ the close of session $t+k$. (a) d1-ultra's top and bottom deciles of the news-implied value V: the level move $\mathrm{CAR}^{\mathrm{lvl}}$ from the previous close, in typical daily moves, with ± 1.96 event-level standard-error bands (descriptive), against each decile's mean V (dashed). (b) The convergence ladder κ_h of Equation (5), with 95% intervals from standard errors clustered by date and firm, against full convergence at $\kappa_h = 1$. d1-ultra is model-specific. The anchor's near-constant readings are in Table 4.

6.2 Convergence to the news-implied value

The convergence analysis asks the same question along the whole path from the open to $t+20$. For each rung h of a ladder from the open to the close of $t+20$, we regress the level move from the previous close to h on the news-implied value:

$$\mathrm{CAR}^{\mathrm{lvl}}_{i,h} = \kappa_h V_i + \alpha_{d(t)\times\tau(i)} + \varepsilon_{i,h}, \qquad h \in \{O_t,\ 10{:}00,\ C_t,\ C_{t+1},\ C_{t+2},\ C_{t+5},\ C_{t+10},\ C_{t+20}\}, \quad (5)$$

where $\mathrm{CAR}^{\mathrm{lvl}}_{i,h}$ is the cumulative abnormal return in typical daily moves, without square-root-of-time scaling, so that it is comparable with V. $\kappa_h = 1$ means that by rung h the price has moved all the way to the model's value; a rising κ_h means drift toward it and a falling one means reversal.

Figure 3 and Table 4 show the ladder. Prices move part of the way to the model's value on day t and no further. For Laya-Fin, κ_h is 0.24 at the open, 0.36 at the close and 0.38 at $t+2$, then falls back to 0.21 at $t+10$ and 0.26 at $t+20$; its pre-registered increase from the open to $t+20$ is 0.02 ($p = 0.47$). For d1-ultra the path is higher and has the same shape: 0.55 at the open, 0.61 at 10:00, 0.59 at the close, then 0.25 at $t+10$ and 0.36 at $t+20$ (increase -0.18, $p = 0.83$). The top decile of d1-ultra's value (mean $V = 0.77$) stands 0.45 above the previous close at the open, 0.52 at the close and 0.36 at $t+20$; the bottom decile (mean $V = -0.64$) stands at -0.40, -0.43 and -0.33. Before the event, prices drift in V's direction: five sessions earlier, the top decile stood -1.19 and the bottom decile 1.54 relative to the previous close. The states describe the company's recent abnormal performance in words, so part of V reflects momentum.

6.3 What the ladder identifies

A slope below one does not by itself mean that the market underreacts, because V is a model's estimate. A standard errors-in-variables argument (Hausman, 2001) makes this precise; we add it

as an interpretation, not as a new test. Suppose the model's value is a noisy measure of the true news-implied value, $V = V^* + e$, with classical error: e has mean zero and is independent of V^* and of the regression error, after the fixed effects are partialled out. Then the slope in Equation (5) converges to

$$\hat{\kappa}_h \;\to\; \theta_V \, \kappa_h^*, \qquad \theta_V = \frac{\mathrm{Var}(V^*)}{\mathrm{Var}(V^*) + \mathrm{Var}(e)} \leq 1, \tag{6}$$

where κ_h^* is the slope on the true value. Two consequences follow. First, the *level* of κ_h mixes market efficiency with model noise, so $\kappa_h < 1$ alone is not evidence of underreaction. Second, the *shape* of the ladder is identified up to the same factor: on a common set of events, $\hat{\kappa}_{h'} - \hat{\kappa}_h \to \theta_V(\kappa_{h'}^* - \kappa_h^*)$. The factor is positive, so the sign is preserved: a rise after the decision point means drift toward the true value and a fall means reversal. This is why the pre-registered tests read the ladder's shape (the increase $\kappa_{t+20} - \kappa_{O_t}$ and the gap closure below) rather than its level, and why the fade after the close is the informative fact in Figure 3: whatever the model's noise, the part of the day's move that lines up with the news does not continue, and partly gives back. The pre-registered increase is not significant for either model, so the fade is descriptive.

Table 4: Convergence to the news-implied value V: prices reach part of V on day t (κ at the close 0.36 for Laya-Fin and 0.59 for d1-ultra), then partly fade; the gap's half-life is not reached within t+20.

	O_t	10:00	C_t	t+1	t+2	t+5	t+10	t+20	Half-life
Panel A: convergence ladder κ_h (D0 events)									
Laya-Fin	0.24	0.27	0.36	0.36	0.38	0.26	0.21	0.26	—
	(0.07)	(0.08)	(0.09)	(0.10)	(0.11)	(0.14)	(0.18)	(0.23)	—
d1-ultra	0.55	0.61	0.59	0.54	0.47	0.36	0.25	0.36	—
	(0.08)	(0.09)	(0.08)	(0.09)	(0.11)	(0.13)	(0.15)	(0.22)	—
anchor	0.08	−0.02	−0.04	−0.04	−0.03	−0.06	−0.03	0.08	—
	(0.08)	(0.08)	(0.07)	(0.09)	(0.09)	(0.10)	(0.11)	(0.16)	—
Panel B: gap closure λ_h and the half-life									
Laya-Fin, D2	—	—	—	−0.007	—	0.026	—	0.050	∞
	—	—	—	(0.012)	—	(0.026)	—	(0.051)	—
Laya-Fin, D1	—	—	−0.011	−0.024	—	−0.002	—	−0.013	∞
	—	—	(0.013)	(0.020)	—	(0.034)	—	(0.062)	—
d1-ultra, D2	—	—	—	−0.009	—	0.022	—	0.049	∞
	—	—	—	(0.012)	—	(0.026)	—	(0.051)	—
d1-ultra, D1	—	—	−0.013	−0.028	—	−0.012	—	−0.020	∞
	—	—	(0.013)	(0.020)	—	(0.034)	—	(0.062)	—
anchor, D2	—	—	—	−0.007	—	0.027	—	0.052	∞
	—	—	—	(0.012)	—	(0.027)	—	(0.051)	—
anchor, D1	—	—	−0.012	−0.025	—	−0.003	—	−0.013	∞
	—	—	(0.013)	(0.020)	—	(0.034)	—	(0.062)	—

Panel A: κ_h is the slope of $\mathrm{CAR}_h^{\mathrm{lvl}}$, the abnormal return from the previous close to rung h in typical daily moves, on V with day × tier fixed effects; $\kappa_h = 1$ is full convergence. Panel B: λ_h is the share of the gap $U = V - G^{\mathrm{lvl}}$ closed by rung h, at D2 (from the close) and D1 (from 10:00); the half-life is the first rung with $\lambda_h \geq 0.5$ (∞: not reached within t+20). Two-way clustered standard errors in parentheses; t+k is the close of session $t + k$. —: not estimated; λ_h is reported at t+1, t+5 and t+20 from D2 and also at C_t from D1, and the half-life belongs to λ_h alone. κ_h to two decimals, λ_h to three, since it is an order smaller.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the pre-registered contamination screens, so their results are model-specific.

6.4 Gap closure

The gap between the model's value and the price after the reaction, $U_{i,\tau} = V_i - G^{\text{lvl}}_{i,\tau}$ at decision point τ (D1 or D2), should close if the price converges to the value. With $\text{AR}(\tau \to h)$ the abnormal move from the decision point to rung h,

$$\frac{\text{AR}_i(\tau \to h)}{\sigma^{cc}_i} = \alpha_{d(t)\times\tau(i)} + \lambda_h U_{i,\tau} + \psi_h V_i + \varepsilon_{i,h}, \qquad h^* = \min\{h : \lambda_h \geq \tfrac{1}{2}\}, \tag{7}$$

where λ_h is the share of the gap closed by rung h, ψ_h captures convergence to a shrunken value, and the half-life h^* is "not reached" when no rung qualifies. The gap does not close. After the close, Laya-Fin's λ_h is -0.007 at $t+1$, 0.026 at $t+5$ ($p = 0.16$) and 0.050 at $t+20$; from 10:00 every λ_h is at or below zero; and the half-life is not reached within $t+20$ for any model. The λ_h are almost the same for every model because V is small next to the day's move, so that $U \approx -G^{\text{lvl}}$: λ_h measures how much of the day's move reverses, not convergence to V. The tracking error between the price and the model's value is no smaller than that of a naive value of zero (ratio 1.001 at $t+20$, $p = 0.87$). None of the convergence tests is supported after Benjamini–Hochberg (Appendix Table A5).

6.5 Under, over and the size of the move

The post-reaction reading classifies each event as an underreaction, a fair reaction or an overreaction, by how far the day's move fell short of or exceeded the model's value. Figure 4 (a) follows d1-ultra's classes after the close, signed in the direction of the news. The 5,021 "Under" events moved against the news on day t (-0.95) and never catch up (-1.04 at $t+20$, against a mean value of 0.38); the 1,783 "Over" events give back part of their move, from 3.16 at the close to 2.82; and the 2,113 "Fair" events stay near 0.70. Laya-Fin calls too few events either way to draw (146 events with $|\hat{N}| \geq 0.5$). Panel (b) asks the question by the size of the move instead. Only the largest moves reverse: moves of at least three typical days reverse slightly in every tercile of news strength (-0.14, -0.18 and -0.09 typical moves over the next week), and smaller moves show small means of both signs. There is no sign that important news with a small move is underpriced.

Two descriptive cuts, pre-registered for reporting but not as tests, show where convergence is strongest. Laya-Fin's slope at the close is 0.75 for the 100 largest companies against 0.23 and 0.25 for the other tiers. And by d1-ultra's own label of the news, prices at $t+20$ reflect the value of numerical news ($\kappa_{t+20} = 0.78$, $p = 0.002$) but not that of narrative news (-0.04).

7 News explains little of a stock's own move; the commercial models are overconfident

7.1 Explaining realized moves

Roll (1988) asked how much of a stock's daily return public news explains, and Kelly et al. (2026) revisit the question for the market with generative AI. We ask it for single stocks, out of sample. For every covered company-day of the test period (days without selected news enter with zero news features, as in Roll), we build up the explanation of the stock-specific move z_{cc} step by step: the market and the sector (Roll's first step), then the pre-open news reading, then the reading of all the day's news, and finally the post-reaction reading's explanation of the move, which reads the realized move and is therefore an attribution, not a forecast. Each step is a ridge regression fitted

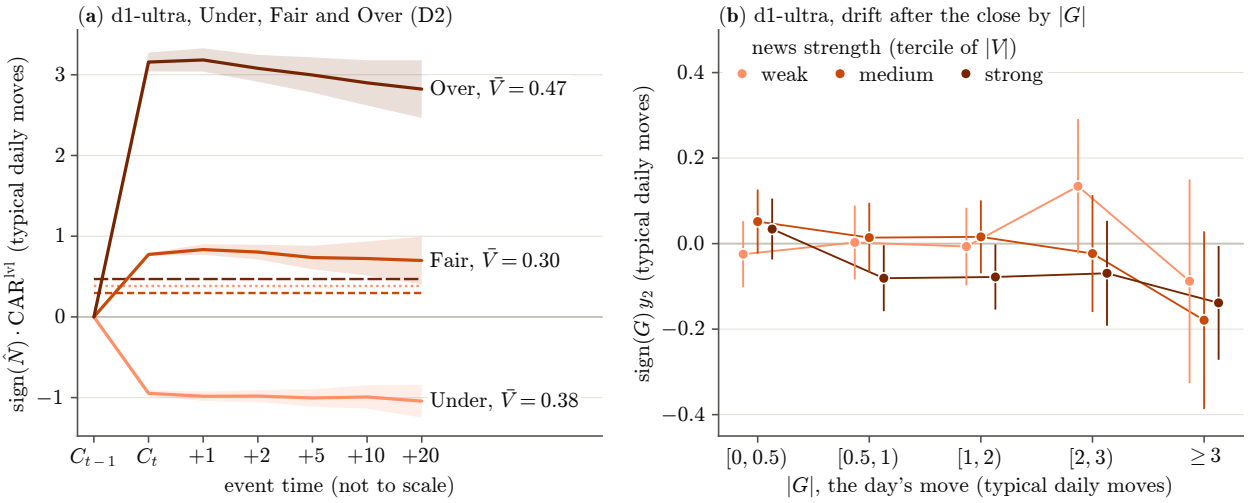


Figure 4: The models' "underreaction" calls never catch up, and only moves of three or more typical days reverse, slightly (−0.09 to −0.18 typical moves over the next week). Test period, d1-ultra (model-specific). (a) D2 events by the post-reaction class, pooled over the size tiers and signed in the news' direction (sign($\hat{N}$) · CAR^{lvl}), with ±1.96 event-level standard-error bands, against each class's mean signed value $\bar{V}$ (dashed: dotted for Under, short dashes for Fair, long dashes for Over). (b) Drift after the close, sign(G) · y_2 (next open to the close of $t + 5$, in the direction of the day's move), by the size of the day's move $|G|$, one line per tercile of news strength $|V|$, with 95% intervals. The strong tercile also reverses slightly for moves of 0.5 to 2 typical days. The pre-registered regression (Table 3) finds no support for H2.

out of sample by the pre-registered protocol (validation data for the fit and the penalty), and each is scored on the test by

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{i,t}(y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t}(y_{i,t} - \bar{y}_{\text{ref}})^2}, \qquad \Delta R^2_s = R^2_{\text{OOS}}(s) - R^2_{\text{OOS}}(s-1), \tag{8}$$

where $\bar{y}_{\text{ref}}$ is the reference the frozen specification fixes (a zero forecast for the stock-specific move, and the val-B mean for the size of the move in H4), $R^2_{\text{OOS}}(s)$ is the fit of steps one to s, and ΔR^2_s is what step s adds to the steps before it. A positive R^2_{OOS} means the step predicts better than the reference on data it was not fitted on.

The news explains little of a stock's own move (Figure 5, Table 5). The market and the sector explain 25.3% of the total daily variance. Of the stock-specific part, the pre-open reading explains 0.13% for Laya-Fin and 0.28% for d1-ultra, and the reading of all the day's news 0.25% and 0.82%. H5, the secondary hypothesis that the same-day news step is positive, is supported for Laya-Fin ($q = 0.058$), the only secondary test that survives Benjamini–Hochberg for it, and for d1-ultra ($q < 0.001$). Much of what the news explains was already knowable before the open (53% for Laya-Fin, 35% for d1-ultra). Reading the realized move explains far more: d1-ultra's explanation step reaches 15.8% of the stock-specific variance on all covered days and 37.5% on days with news.

Being explained does not predict persistence. Days the model judges explained (the probability that the move is explained by the news is at least one half) are 16% of event days but carry 41% of their stock-specific variance. Yet in the pre-registered persistence regression the coefficient on the explanation is −0.062 after the close (standard error 0.038, $p = 0.95$ for a positive effect): explained moves do not persist more than unexplained ones, and if anything reverse slightly more. The size of the move is easier to read than its direction, as Section 4 showed for H4: the models rank which stocks will move a lot, but add nothing to recent volatility.

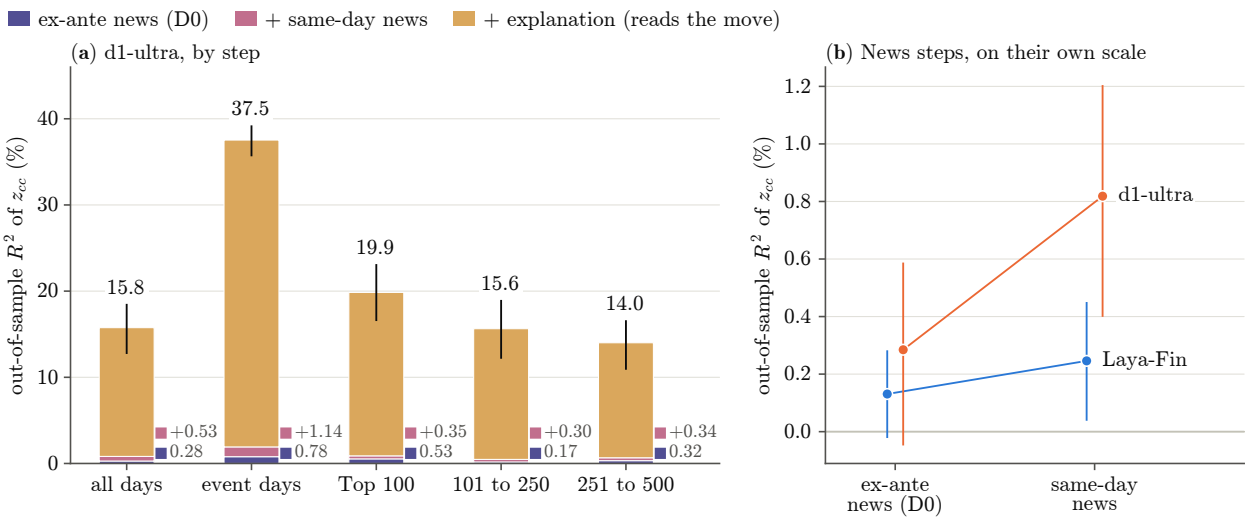


Figure 5: Same-day news explains 0.25% of the stock-specific daily variance out of sample for Laya-Fin and 0.82% for d1-ultra; reading the realized move explains 15.8%. Test period; out-of-sample R^2 in percent of the stock-specific variance (the market and the sector removed). (a) d1-ultra's R^2 by cut, stacked by step in the legend's order: the pre-open news, plus the day's news, plus the post-reaction explanation (which reads the realized move, so it is not a forecast). Each total stands above its bar with its 95% bootstrap interval; the two thin news steps are labelled beside each bar. For comparison, the market and the sector explain 25.3% of the total daily variance, a different base. (b) The news steps on their own scale, with 95% bootstrap intervals, for Laya-Fin and d1-ultra (model-specific). Only d1-ultra answers the explanation question; the anchor's near-constant readings are in Table 5.

Table 5: Explaining realized moves: same-day news explains 0.25% of the stock-specific variance for Laya-Fin (H5, supported) and 0.82% for d1-ultra; reading the move explains 15.8%.

	Laya-Fin		d1-ultra		Anchor	
Step	R^2	ΔR^2	R^2	ΔR^2	R^2	ΔR^2
Panel A: the explanation ladder, out-of-sample R^2 of z_{cc} (%)						
+ ex-ante news (D0)	0.13	—	0.28	—	0.01	—
+ same-day news	0.25	0.12	0.82	0.53	0.03	0.02
+ explanation (reads the move)	—	—	15.76	14.94	—	—
share knowable before the open (%)	53.17	—	34.78	—	35.63	—
Panel B: the secondary test H5						
H5: same-day news step, $R^2 > 0$	0.25	—	0.82	—	0.03	—
one-sided p	0.010	—	<0.001	—	0.072	—
BH q	0.058	—	<0.001	—	0.358	—

Covered member-days of the test period; out-of-sample R^2 of the stock-specific move z_{cc} (the market and sector removed) by cumulative step, in %; ΔR^2 is the step's increment. Roll's first step, the market and the sector, explains 25.3% of total daily variance, a different base. The explanation step needs the post-reaction reading's explanation answer, which only d1-ultra gives; it reads the realized move, so it is not a prediction. H5 is the only secondary test that survives Benjamini–Hochberg (q 0.10) for Laya-Fin. —: not applicable (the first step has no increment; only d1-ultra answers the explanation question; the share and Panel B have one value per model). R^2, ΔR^2 and the share in % to two decimals; p and q to three.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the pre-registered contamination screens, so their results are model-specific.

7.2 Calibration

The pre-registered calibration metric is the expected calibration error (Pakdaman Naeini et al., 2015; Guo et al., 2017): the pre-open reading's nine levels are grouped into three classes (a fall, no move and a rise, split at half a typical daily move), and the top class's stated probability is compared with its hit rate against the realized five-day move, in 15 equal-width bins. Laya-Fin is calibrated but compressed: its error is 0.057 and its log-loss improves on the validation base rates by only 0.001, because its probabilities stay close to those base rates. The commercial models are overconfident: their stated probabilities overstate their hit rates, with errors of 0.260 (d1-ultra), 0.361 (span-02) and 0.209 (drex-v2.0) and negative log-loss skill. A single temperature fitted on val-A lowers d1-ultra's error from 0.260 to 0.155 (Appendix Table A6 and Figure A2). Fine-tuning does not buy a significant IC gain over the untuned model either: H7's difference between Laya-Fin and Laya zero-shot is 0.015 ($p = 0.097$, $q = 0.24$). The cost per thousand decisions is zero for the trained models on a free GPU tier, and 0.028 to 0.063 dollars for the commercial models.

8 What the models know about masked firms

Masking removes names, but news is distinctive: a drug approval, a store count or a guidance range can point to one company. We measure how often a model recognizes the company behind a masked state. Each of 400 masked val-A states is shown with four candidate companies from the same sector and size tier, the true one and three distractors chosen to match its news activity, and the model picks one. For model m,

$$\mathrm{Acc}_m = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\left[\hat{c}_i = c_i\right], \qquad \mathrm{excess}_m = \mathrm{Acc}_m - p_0, \tag{9}$$

where c_i is the true company, $\hat{c}_i$ the model's choice and $p_0 = 0.28$ the matched baseline: the accuracy of a guess that uses no reading, only the candidates' news activity. Intervals are Wilson score intervals (Wilson, 1927). The matched baseline matters: with distractors drawn at random, a guess based on news activity alone already scores 36%, because the true company always has news.

The commercial decision models recognize the company far more often than the baseline (Figure 6): from 62% of states (mercury-decide) to 74% (d1-ultra), and 66% and 68% for the other two comparators. The open-weights Laya model, the base of Laya-Fin, does so in 32%, close to the baseline. All of the commercial models therefore fail the pre-registered screen of 40%, and every model in the figure was released after the sample ended: Laya on 2026-09-18 and the commercial models between 2026-10-12 and 2026-10-16. Their training data are undisclosed, so we cannot rule out that they saw text from the test period.

Recognition is not the same as remembering returns, and two results suggest that it did not drive the commercial models' signal: unmasking the states does not raise d1-ultra's IC (Section 5), and their ICs fell from validation to test as the other signals' did, which is not what an advantage from having seen the test period would produce. Neither is a test of memorization. The result is, however, a warning for backtests of language models on historical news: masking hides names, not companies. The pre-registered forward test (Section 11) asks the question directly, on news published after the models were released.

9 Economic significance

The pre-registered strategies turn each reading into a daily long-short book within size tiers, hedged against the market with SPY, and ask whether the readings are worth trading. The *convergence*

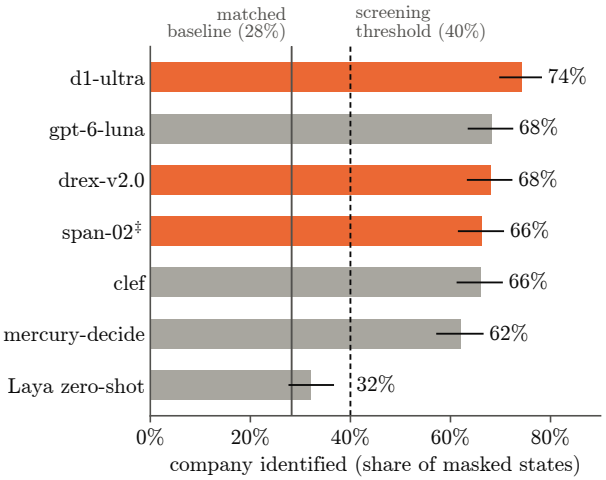


Figure 6: Commercial decision models identify the company behind masked news in 62% to 74% of states, against a matched baseline of 28%; the open-weights Laya model does so in 32%. Company identification on masked val-A states with activity-matched distractors: the share of states in which each model names the company among four candidates, sorted, with its Wilson 95% interval, against the matched baseline (a guess that uses no reading; solid line) and the pre-registered screening threshold (40%; dashed line). The three commercial models carried to the test are in the comparator colour, the other commercial models and Laya zero-shot in gray. Every model was released after the sample ended. Identifying a company is not the same as remembering its returns. ‡span-02: see Figure 1.

strategy buys the events whose price lies furthest below the model's value (the top 20% of the gap U) and sells those furthest above it, after the first half hour (D1) or after the close (D2). Three more families rank the pre-open reading (entering at the open), and the post-reaction judgment J, signed by the move, at 10:00 and after the close. The tables label the books by their entry point: S-D0 for the pre-open reading, S-D1 and S-D2 for the judgment at 10:00 and after the close, and S-CONV for the convergence strategy; an RG book trades only the events the model calls an under- or over-reaction, and a gated book only its most confident readings. Costs are charged at 5, 10 and 20 basis points per side on both legs and on the hedge. Each book is compared with 100 books built on random signals over the same events, and its Sharpe ratio is deflated for the number of trials behind it (Bailey and López de Prado, 2014). We deflate with the 545 trials the ledger held when the strategies were evaluated: every selection of filters, encodings, models and baselines, and every strategy variant up to then. The exploratory study of Section 10 later raised the ledger to 1,620 trials, which would only raise the bar (Appendix C gives the formula).

No pre-registered strategy is worth trading after costs (Table 6 and Figure 7). Gross of costs, no book reaches the 95th percentile of its random-signal placebo, and Laya-Fin's post-reaction book after the close falls below the 5th (percentile 3), as expected of judgments that carry no positive information about the next week. Net of 10 basis points every main book loses except the tiny book on the model's under- and over-reaction calls, which holds one stock a side on a handful of days. The convergence strategy earns a gross Sharpe ratio of 0.61 for Laya-Fin and -1.98 net of costs. It is, in effect, a price-only gap fade: because the model's value is small next to the day's move, ranking by the gap $U = V - G^{\text{lvl}}$ buys the day's losers and sells its winners. The two books share 92% of their positions, 99.8% of them on the same side, and their daily returns correlate 0.96. The highest gross Sharpe ratio, 1.88 for the pre-open book restricted to Laya-Fin's most confident readings and held to the next close, reaches percentile 93 of its placebo and breaks even at 8.3 basis points per side. The one-day books turn over about 4.4 times their capital a day, so costs decide

their sign. With 545 trials the deflation benchmark is a daily Sharpe ratio of 0.26, and the largest deflated-Sharpe probability of any test book is 0.024. Short-sale borrow costs, which would lower every net figure, are not modelled.

Table 6: Strategies on the test period: no pre-registered book survives costs and the deflated Sharpe ratio (the largest deflated-Sharpe probability is 0.024).

		Sharpe ratio					
Book	Holding	Gross	Net	Turnover	Break-even (bp)	Placebo pct.	DSR
Laya-Fin (primary)							
S-CONV	D2 to t+5	0.61	−1.98	0.90	2.4	68	<0.001
S-D0	open to close	0.35	−6.03	4.36	0.5	64	<0.001
S-D0, gated	open to t+1	1.88	−0.38	2.31	8.3	93	<0.001
S-D1 J	10:00 to close	0.76	−9.70	4.36	0.7	71	<0.001
S-D2 J	next open to t+5	−2.69	−6.45	0.83	−7.1	3	<0.001
S-D2 RG	next open to t+5	1.69	1.50	0.05	85.1	92	0.024
d1-ultra (model-specific)							
S-CONV	D2 to t+5	−0.09	−2.72	0.90	−0.3	48	<0.001
S-D0	open to close	0.10	−6.41	4.41	0.1	58	<0.001
S-D1 J	10:00 to close	−0.28	−11.35	4.51	−0.3	40	<0.001
S-D2 J	next open to t+5	0.39	−2.37	0.85	1.4	64	<0.001
Benchmarks							
gap fade (price-only)	D2 to t+5	0.44	−2.01	0.90	1.8	66	<0.001
vendor sentiment†	open to close	0.27	−6.24	4.36	0.4	62	<0.001

Annualized Sharpe ratios ($\sqrt{252}$), gross and net of 10 bp per side on both legs and the SPY hedge; no borrow cost. Turnover: dollars traded a day per dollar of capital. Break-even: the cost per side, in basis points, at which the mean net return is zero (gross mean ÷ turnover). Placebo: the percentile of the gross Sharpe among 100 random-signal books on the same events. DSR: the deflated-Sharpe probability at 10 bp with $N = 545$ trials, every trial logged when these pre-registered strategies were evaluated (the ledger holds 1,620 after the post hoc exploratory study). Each family is shown at its hypothesis' horizon; every book is in the replication package. S-CONV ranks by the gap $U = V - G^{\text{lvl}}$; S-D0 by $\tilde{N}$; S-D1 and S-D2 by the judgment J signed by the move; RG books trade the cases the model calls a misreaction (one name a side).

†Vendor sentiment: its labels were produced retrospectively, so its books are a possibly look-ahead-contaminated upper bound, not a benchmark; the clean benchmarks are the price-only gap fade and the random-signal placebo. The API models fail the pre-registered contamination screens, so their results are model-specific.

10 Exploratory analyses (post hoc): where the signal lives

Everything in this section is exploratory. The analyses were requested after unblinding, they are outside every pre-registered family and they support no claim; they suggest where to look, and the forward test of Section 11 re-tests two of them as pre-registered hypotheses. Earlier looks at the test period by the authors' AI research assistant, at two of the statistics below (hit rates and mean moves by conviction band, and the statistics of the 10:00 reading), shaped the design of this exploratory study. Every exploratory variant is logged as a trial: the ledger grew from 545 to 1,620 trials.

Earnings days carry the signal. In a post hoc split, every signal's pre-open IC is concentrated on the events with a scheduled earnings report (1,128 of the 13,748 test events), and there it held in all three periods. In this exploratory split, Laya-Fin's IC with z_{cc} on earnings days was 0.125,

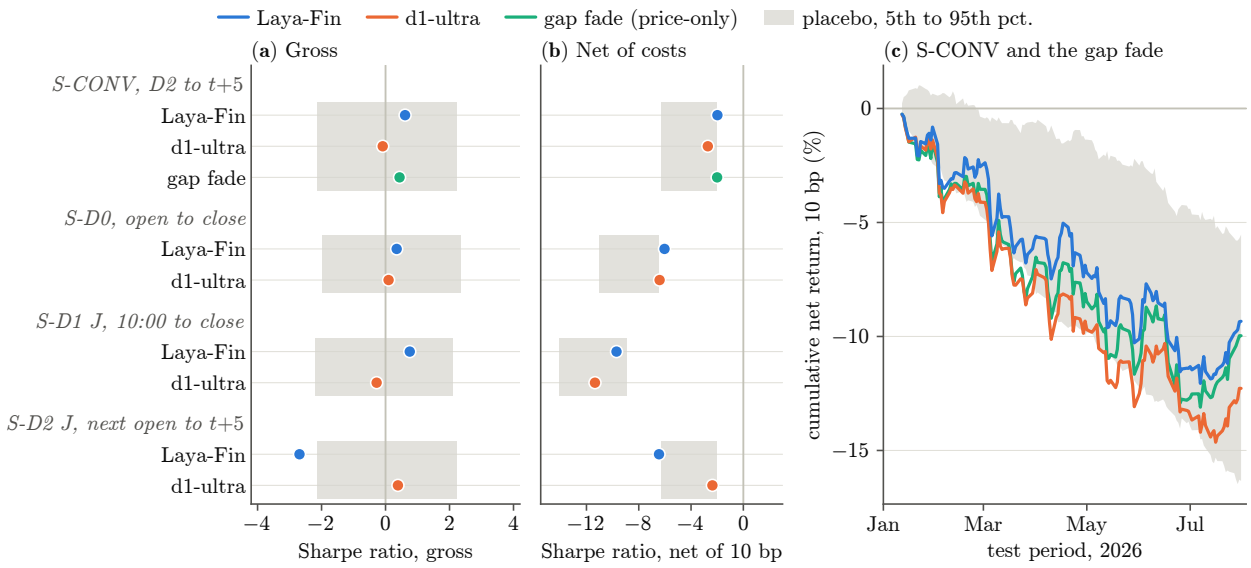


Figure 7: No pre-registered strategy survives costs: net of 10 basis points every main book loses, and the convergence strategy is effectively a price-only gap fade (92% of positions shared, daily correlation 0.96). Test period. (a, b) Annualized Sharpe ratio of each main book, gross and net of 10 basis points per side on both legs and the SPY hedge, at the exit of its hypothesis' horizon; the gray box is the 5th to 95th percentile of 100 random-signal books on the same events; (b) has its own scale. (c) The convergence strategy's cumulative net return (Laya-Fin and d1-ultra) against the price-only gap fade, with the placebo's 5th to 95th percentile band. Every book is in the replication package. d1-ultra is model-specific.

0.106 and 0.134 on val-A, val-B and the test ($t = 3.2$ on the test), against 0.048, 0.049 and 0.013 on the other events, and d1-ultra's was 0.304, 0.237 and 0.223 ($t = 8.1$), against 0.057, 0.062 and 0.018. Post hoc, the fall from validation to test of Section 4 is therefore a fall on the other events. The exploratory split also shows the commercial models pricing earnings news at the open: d1-ultra's earnings-day IC on the test is 0.238 with z_{on} and 0.034 with z_{id}.

Selective prediction: the most confident calls carry most of the signal, in the overnight gap. We ranked events post hoc by conviction, the size of $|\hat{N}|$ against its val-A distribution, and kept only the top share of them, which traces a coverage curve (Geifman and El-Yaniv, 2017) (Figure 8). In this exploratory analysis, d1-ultra's mean overnight move in the direction of its reading, $\text{sign}(\hat{N}) \cdot z_{on}$, is 1.52, 1.16 and 0.93 typical moves in its top 10% (val-A, val-B and the test), against 0.22, 0.26 and 0.21 for all events; for the close-to-close move the post hoc figures are 0.80, 0.75 and 0.59 against 0.13, 0.16 and 0.10. Post hoc hit rates move less: d1-ultra's top 10% gets the sign of z_{cc} right 0.556 of the time on the test, against 0.506 for all events. After the open, selectivity gains little in this exploratory analysis: the top 10%'s mean move in the direction of the reading is 0.101 for d1-ultra (95% interval −0.021 to 0.241) and 0.068 for Laya-Fin on the test, and no test interval of this post hoc mean excludes zero at any coverage.

Watching the open adds nothing. In a post hoc comparison, the reading at 10:00, after the first half-hour bar, never predicts the rest of the day better than the pre-open reading on val-B or the test (the one exploratory exception is d1-ultra on val-A, 0.029 against 0.014, neither significant). Post hoc, the revision from the pre-open to the 10:00 reading has a zero or negative IC with the move from 10:00 to the close on val-B and the test (Laya-Fin −0.027 and −0.036, $t = -2.3$ on the

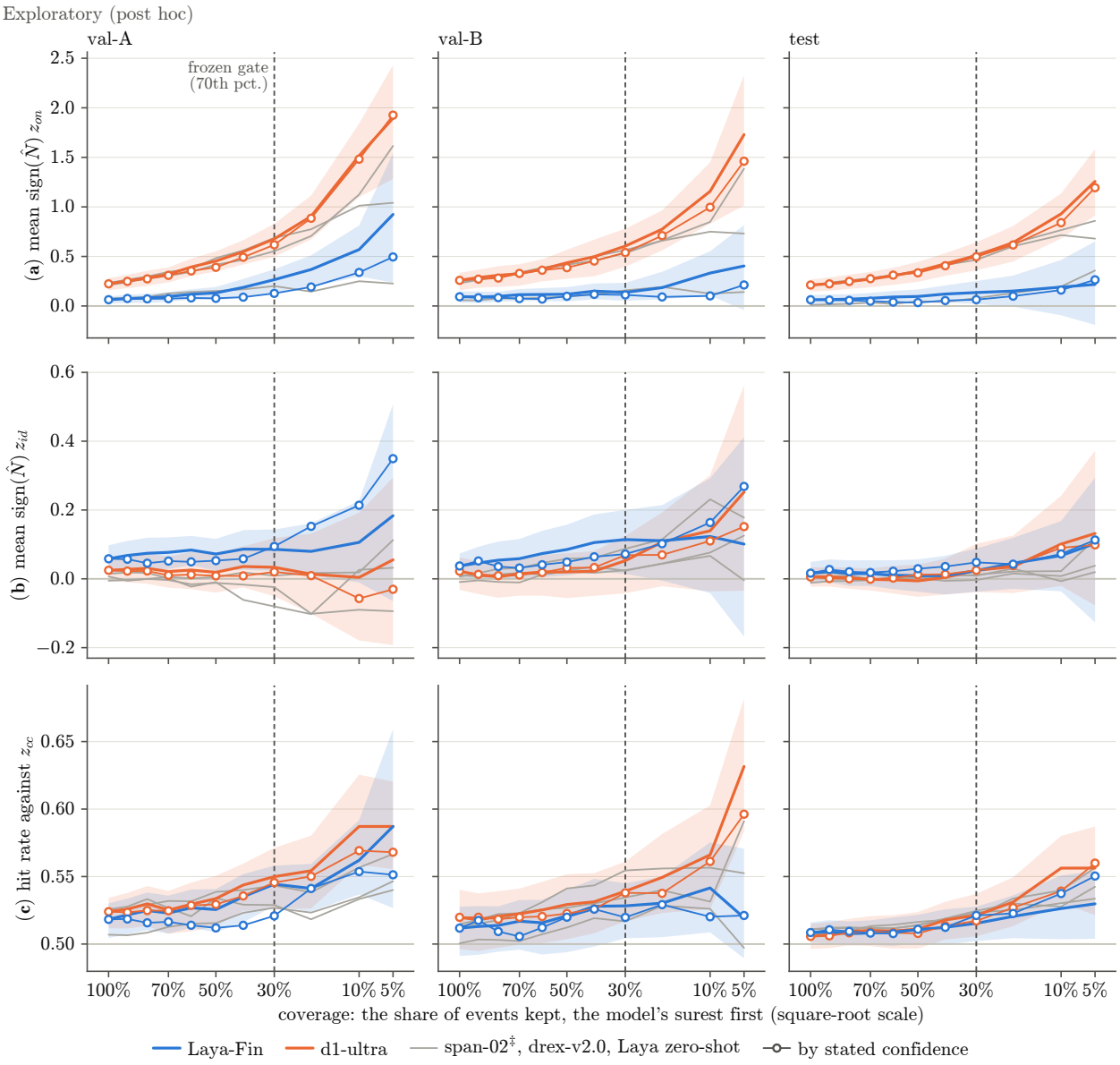


Figure 8: Exploratory (post hoc): d1-ultra's most confident tenth of pre-open readings carries most of the overnight signal in all three periods (mean signed overnight move 1.52, 1.16 and 0.93 typical moves, against 0.22, 0.26 and 0.21 for all events), but not the move after the open. Coverage curves: events ranked by conviction $|\hat{N}|$ (lines) or by the stated confidence (circles), keeping the most confident share, with thresholds from val-A; 95% intervals from the stationary bootstrap over sessions. (a) Mean sign$(\hat{N}) \cdot z_{on}$; (b) mean sign$(\hat{N}) \cdot z_{id}$; (c) the hit rate against z_{cc}. The dashed line marks the frozen confidence gate (the val-A 70th percentile). Requested after unblinding; the authors' assistant had looked at the conviction bands on the test before this study, which shaped it. d1-ultra, span-02‡ and drex-v2.0 are model-specific.

test; d1-ultra -0.043 and -0.011): in this exploratory comparison the models over-update toward the first bar. The comparison involves many statistics, so a few large t-values are expected by chance and the post hoc pattern is weak evidence.

Validation-to-test decay, the strongest instance. A post hoc conditioning study asked whether the after-the-open signal $\text{sign}(\hat{N}) \cdot z_{id}$ is stronger under some condition: the type and timing of the news, d1-ultra's own labels of it, the day, the stock, the models' internals, agreement between models and a cost hurdle, for 75 (model, condition) pairs. A strict rule of this exploratory study, fixed before its test run, selected the pairs positive with $t \geq 1$ in both validation quarters and a pooled $t \geq 2$: 8 pairs passed, with pooled t from 2.3 to 3.5. On the test, 5 of the post hoc selections changed sign and the other 3 are small (largest $t = 1.01$); in this exploratory family the smallest Benjamini–Hochberg q is 0.94 among the selected pairs and 0.79 overall (Appendix Figure A3). The exploratory tradable versions had in-sample gross Sharpe ratios of 4.60 to 6.78 (Laya-Fin) and 1.69 to 2.25 (d1-ultra) on val-B; on the test these post hoc books' gross Sharpe ratios run from -0.73 to 0.67, none reaches the 90th percentile of its placebo, and every one loses net of 10 basis points. A post hoc moving-average-cross overlay chosen on validation failed the same way: after crosses of the 20- and 50-day averages without news, the mean abnormal return in the direction of the cross over the next five days was -41.1 basis points on validation ($t = -2.1$) and -14.4 on the test ($t = -0.7$), so this post hoc reversal did not replicate.

11 Limitations and the forward test

Limitations. The sample is the S&P 500, the largest and most closely followed US stocks; smaller stocks, where news may diffuse more slowly (Hong et al., 2000), may behave differently. The horizon is daily, with one intraday decision point at 10:00 built from 30-minute bars whose first bar starts at the first trade, not at the opening auction. The test period is seven months long (140 sessions with D0 events), so small ICs are hard to detect, and the selection rested on two validation quarters of 59 sessions each. The trained instruments were trained once, with one seed and one epoch, on a free GPU budget; the anchor learned the size of moves but not their direction and is uninformative, so the study says nothing about what fine-tuning adds. The commercial models failed the pre-registered screens and were released after the sample, so their results are model-specific and contamination cannot be ruled out. The vendor-label baselines rest on labels produced after the fact. Masking leaves small residues of names and dates, below half a percent of states. The strategies charge flat costs per side and model no borrow costs or market impact, and the hedge removes market risk only. Some pre-registered items were not run or not completed: three robustness variants that needed new model calls (a summary-text variant, vendor tags in the text and a 09:30 news cutoff; the 08:00 cutoff is approximated), the second and third training seeds, and 210 rate-limited requests of the ablations, which were not retried.

The forward test. The cleanest test of memorization uses news published after every model's release. Seven hypotheses for such a test were pre-registered on 2026-10-29, before any news from the forward window was exported and before any forward-period model call; the window runs from the first session after 2026-10-21 for at least three months, including the January–February 2027 earnings season, with the frozen pipeline run unchanged. They are:

- **F1, memorization:** for each commercial model, the change in IC from the test to the forward window minus the same change for the clean baselines is negative (Holm across the three models);

- **F2, entity against event knowledge:** company identification on masked forward states against masked test states, and against the matched baseline;
- **F3, the conviction book:** the pre-open book restricted to each model's top tenth of $|\hat{N}|$ (thresholds fixed from val-A) has a positive net Sharpe ratio;
- **F4, priced at the open:** the frozen H1 and H1b on the forward window;
- **F5, over-updating:** the IC of the revision after the first half hour with the rest of the day's move (two-sided);
- **F6, overconfidence:** the commercial models' expected calibration error exceeds Laya-Fin's (Holm across the three models);
- **F7, conviction concentration:** the mean signed overnight move of the top tenth by conviction exceeds that of all events (Holm across four models).

F5, F6 and F7 turn the exploratory results of Sections 7.2 and 10 into confirmatory tests. The forward test has not been run. The robustness runs of the anchor (seeds 17 and 23 and a longer training run), fixed before unblinding, have not been run either; they are robustness only and cannot change any verdict here.

12 Conclusion

AI decision models read financial news well enough to forecast how a stock will open, and not well enough to forecast what it does after that. A fine-tuned open-weights model that looked convincing on two validation quarters did not survive a blinded test; the commercial models kept a signal, but only in the overnight gap, where the market had already priced the news before anyone could trade on it at the open, as an efficient market would (Fama, 1970). The models could not tell an overreaction from an underreaction, their calls of underreaction never caught up, and only the largest moves reversed, slightly. No pre-registered strategy survived costs and selection. Out of sample, news explains a fraction of a percent of a stock's own daily variance, and the commercial models' stated probabilities overstate their accuracy.

Two lessons reach beyond these models. First, validation on a few quarters is weak evidence: the strongest exploratory pattern we found, with in-sample Sharpe ratios up to 6.8, reversed on the test. A frozen plan, a trial ledger and a single unblinding are what made that visible. Second, masking is not anonymity: commercial models recognize most masked firms, so backtests of language models on historical news should measure recognition directly and, where possible, test on news published after the model's release, as our pre-registered forward test will. The market reads the news first.

Acknowledgments and statements

[Acknowledgments, funding and conflicts of interest, to be provided by the authors.]
[Statement on the use of AI tools, to be confirmed by the authors: an AI coding and research assistant implemented and ran the pre-registered pipeline under the authors' direction, and drafted this manuscript from the verified outputs; the authors are responsible for the content.]

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A Pre-registration and deviations

The pre-registration fixes every definition, constant, split, hypothesis and selection rule. After it was written, changes could only be appended to its deviations log, each with its date, the section it touches, the reason and whether it came before or after the freeze; the freeze record holds the log's length at the freeze, and the lock refused to release the test outcomes unless the log was an unchanged prefix of the current one. Table A1 gives every entry in one line. Most entries make precise what the pre-registration left open (how a filter reaches its budget, which events a sample holds, how a variable is scaled) and were written before the freeze. The material deviations are these.

- *Amendments before any data.* Convergence to the news-implied value, a wider question battery and the firm-level explanation analysis were added on the day the pre-registration was written.
- *Masking.* The masking rules were widened several times before the freeze, after independent checks found residual names, dates and price moves; non-English articles, move explainers and running market stories were excluded; residues below half a percent of states were accepted and disclosed.
- *The commercial models* failed the pre-registered screens. Instead of dropping them, the three with the highest second-stage IC were kept as screen-failing, model-specific comparators (before the freeze).
- *Training.* The trained instruments are one seed and one epoch each, not the pre-registered three-seed ensemble, because of a week's free GPU quota; the other seeds are robustness runs, fixed before unblinding and not yet run.
- *H1b* (the IC with the move after the open) was added to the primary family before the freeze, because the strategy that enters at the open needs it.
- *Company identification* was re-run with activity-matched distractors before the freeze, because a guess from news activity alone beat chance in the original design.
- *Items not run.* Three robustness variants needed new model calls and were not run (a summary-text variant, vendor tags in the text and a 09:30 news cutoff); the 08:00 cutoff is approximated from time stamps. Of the ablation requests, 210 were rate-limited and not retried.
- *The canary.* After the freeze, a stability check that repeats earlier requests fired for span-02 before production. Its build string was unchanged and the drift was close to its own repeat noise, so production went ahead under a rule recorded before the end-of-production check: a model counts as updated only if its build changes or two independent checks fire. At the end of production neither condition held for any model.
- *The price-only baseline's post-close predictions* were moved after unblinding, because one of their inputs, the day's overnight move, is a test-period outcome; the model was unchanged and first reproduced its validation predictions exactly.
- *Analysis and strategies.* The operationalization of the frozen specification and of the strategies was written before the outcomes were read; three code fixes found when the code first met the test data changed no pre-registered number.
- *Exploratory work after unblinding* is dated in the log. The conditioning study's validation selection was committed before its single test run; its two definitional choices (the VIX close of the previous session, read from a registered copy of the market data, and payroll release dates added from the Bureau of Labor Statistics) were made on validation data.

- *The forward test* was pre-registered after the paper's analyses and before any forward-period data existed (Section 11).

Table A1: Deviations from the pre-registration: all 66 entries of its deviations log, one line each.

Row	Date	Stage	Change	Reason	Freeze
1	10-24	setup	Freeze hash stops at the log; the log is append-only	The log must grow after the freeze	before
2	10-24	setup	Test-year adjusted closes sit outside the sealed folder	Raw data are read-only	before
3	10-24	operator	Amendment: convergence to V and its strategy	Operator, before any data	before
4	10-24	operator	Amendments: question battery v2, explaining moves	Operator, before any data	before
5	10-24	prices	Share classes grouped by issuer name	The database rarely has a CIK	before
6	10-24	prices	Company cap from one class; shares on the split basis	Vendor shares are company-level	before
7	10-24	prices	Test-month ranking closes rebuilt from adjusted ones	Tiers needed before the freeze	before
8	10-24	prices	Stocks without adjusted prices use total returns	Else about 7% of members drop	before
9	10-24	prices	Sector pool: one class per company, own company out	No company in its own factor	before
10	10-24	prices	Target details: volume, σ floor, multi-day scaling	The plan leaves them open	before
11	10-24	price checks	Share counts checked against a second source	Bad vendor shares for large names	before
12	10-24	price checks	Price-level check at dividend dates; one stock rescaled	One stock's closes were halved	before
13	10-24	price checks	Raw volume put on the traded share basis	Pre-split volume was adjusted	before
14	10-24	news	News tickers mapped through dated aliases	The database holds no renames	before
15	10-24	news	Near-duplicates grouped in time order; novelty fixed	Keeps grouping point-in-time	before
16	10-24	news	Rules of the after-the-fact price-talk detector	The plan names categories only	before
17	10-24	news	Leakage audit of vendor sentiment (z_{id} vs z_{on})	The plan states it in words	before
18	10-25	filters	Rule filters reach the budget by seeded thinning	A yes-or-no rule has no threshold	before
19	10-25	filters	Filter comparison sample purged and embargoed	Purge and embargo left open	before
20	10-25	filters	Learned filters trained on articles; tuning metric fixed	Training unit left open	before
21	10-25	filters	Post-close events, third-party exclusion, data end	The plan does not cover them	before
22	10-25	masking	The company block carries the sector only	The database has no industry	before
23	10-25	masking	Codebook scales for multi-day moves, market series	The plan gives them no scale	before
24	10-25	masking	Publication lags of market, macro and earnings fields	Availability left open	before
25	10-25	masking	Masking widened: periods, dates, tickers, names	Residue above the target	before
26	10-25	masking	No pre-market talk in post-reaction states	Numbers stay in the codebook	before
27	10-25	masking	Move explainers and aggregator reposts left out	They carry the session's move	before
28	10-25	masking	Residual names (about 1% of states) accepted	Content alone identifies half	before
29	10-26	masking	Non-English articles and English price talk removed	Price talk in 2.5% of states	before
30	10-26	masking	More date forms; every price move and level removed	Date residue failed a check	before
31	10-26	masking	Running market stories left out of pre-open states	Live pages held the outcome	before
32	10-26	masking	Residues below 0.5% of states accepted, disclosed	The final check passed	before
33	10-26	news	Articles of undecided parsing status allowed	Operator decision	before
34	10-26	tournament	Tournament samples, IC, confidence set, screens fixed	The plan leaves details open	before
35	10-27	operator	No API model passes the screens; top three kept	Within-model tests stay valid	before
36	10-27	encoding	Encoding search and anchoring test operationalized	The plan gives rules only	before
37	10-27	encoding	Numeracy gate for encodings other than the default	The fallback was undefined	before
38	10-27	operator	Intraday bar request cap raised from 3,000 to 17,000	The vendor serves 30 days a call	before
39	10-27	bars	Decision point D1: the 10:00 price, returns, scales	The plan leaves them open	before
40	10-27	bars	Per-session basis of 30-minute bars; 5 bp bound failing	Vendor data need it	before
41	10-27	bars	The D1 post-reaction state and its half-hour wording	No phrase for the 10:00 move	before
42	10-27	bars check	Request counts and the 5 bp gap corrected	Recomputed from the ledger	before
43	10-27	operator	Two attention cuts; timing and text robustness	Limited attention; leak checks	before
44	10-26	training	Trained models read 1,036 tokens instead of 1,024	No state or question is cut	before
45	10-27	training	One seed and one epoch per trained model for now	One week's free GPU quota	before
46	10-27	baselines	Baselines operationalized: events, features, LLM	The plan names them only	before
47	10-28	freeze	Analysis spec: samples, inference, bootstrap, families	Frozen code must fix them	before
48	10-28	freeze	Freeze hash also covers inference and metrics code	The frozen spec runs them	before
49	10-28	operator	Trained instruments pinned to the seed-13 runs	Later seeds must not change them	before
50	10-28	freeze	Final selection: val-B IC on common events	Event set left open	before
51	10-28	operator	Company identification with matched distractors	Removes baseline bias	before
52	10-28	freeze	Live-format residue counted with the frozen pattern	First count was narrow	before
53	10-28	operator	H1b, the IC with z_{id}, joins the Holm family	Tests what the open book trades	before
54	10-28	operator	Production plan: d1-ultra as the API comparator	The specification names no rule	before
55	10-28	operator	Three robustness variants dropped; 08:00 approximated	They need new states	before
56	10-28	operator	Canary baselines and drift check for API models	None existed for them	before

continued on the next page

Table A1, continued

Row	Date	Stage	Change	Reason	Freeze
57	10-28	operator	span-02's canary stop judged noise; proceed	Inside its own repeat range	after
58	10-28	operator	Price-only post-close test predictions after unblinding	A feature is a test outcome	after
59	10-28	operator	Anchor robustness runs fixed before unblinding	Near-constant anchor outputs	after
60	10-28	analysis	Calls of the frozen spec fixed before unblinding	Caller fixes joins, families	after
61	10-28	analysis	Figure data definitions and code fixes	No pre-registered number moves	after
62	10-28	operator	Exploratory analyses after unblinding, dated	Paper trail for post hoc work	after
63	10-28	strategies	Strategy engine fixed before any test-period return	The plan gives books, not joins	after
64	10-29	operator	Forward test pre-registered: F1 to F5	Confirmatory forward test	after
65	10-29	exploratory	Exploratory study: rules fixed before its test	The request leaves joins open	after
66	10-29	operator	Forward test: F6 overconfidence, F7 conviction	Re-tests exploratory findings	after

Dates are in the sample's last calendar year (MM-DD); Stage: the part of the project that wrote the entry; Freeze: written before or after the freeze. The summaries are the authors'; the full text of each entry is in the pre-registration's deviations log, in the replication package.

B Notation

Table A2: Notation.

Symbol	Meaning
D0, D1, D2	Decision points: 09:15 ET before the open (pre-open reading); 10:00 ET after the first 30-minute bar and after the close (post-reaction readings).
val-A, val-B, test	2025-07-01 to 2025-09-30; 2025-10-01 to 2025-12-31; 2026-01-02 to 2026-07-31.
r^{cc}, r^{on}, r^{id}	Close-to-close, overnight (close to open) and intraday (open to close) returns, with $1 + r^{cc} = (1 + r^{on})(1 + r^{id})$; Equation (1).
AR^k, $\sigma^k_{i,t-1}$	Abnormal return (market and leave-one-out sector removed, betas over 252 sessions to $t-1$); its ex-ante scale (1.4826 times the median absolute deviation over 126 sessions to $t-1$, floored at the day's cross-sectional 10th percentile).
z^k; z_{cc}, z_{on}, z_{id}	$\mathrm{AR}^k/\sigma^k_{i,t-1}$, the move in typical daily moves (clipped at ± 10).
y_0, y_1, y_2	Outcomes after D0, D1 and D2: z_{id}; the move from 10:00 to the close; the move from the next open to the close of $t+5$, each over its ex-ante volatility.
e_5	The move from the previous close to the close of $t+4$ over $\sigma^{cc}\sqrt{5}$: the training label and the calibration target.
G, G^{lvl}	The observed reaction at D1 or D2 in typical-move units; its level version, divided by σ^{cc} without a square-root-of-time factor.
$\hat{N}$, $\tilde{N}$	The news-implied move $\sum_k p_k m_k$ over the nine levels of the pre-open reading; its calibrated version (an isotonic map to e_5 fitted on val-A), which ranks the pre-open strategy.
V, U	The news-implied value (an isotonic map of $\hat{N}$ to the level move to $t+4$, fitted on val-A); the remaining gap $V - G^{\mathrm{lvl}}$ at D1 or D2.
J	The post-reaction judgment $P(\text{much too small}) + P(\text{too small}) - P(\text{too large}) - P(\text{much too large})$.
Under, Fair, Over	Post-reaction classes from $(\hat{N} - G)\,\mathrm{sign}(\hat{N})$ for $\lvert\hat{N}\rvert \geq 0.5$: Under above 0.5, Over below -0.5, Fair between.
$\mathrm{CAR}^{\mathrm{lvl}}_h$	The level cumulative abnormal return from the previous close to rung h.
κ_h, θ_V	The convergence slope at rung h (Equation (5)); the reliability of V (Equation (6)).
λ_h, ψ_h, h^*	The share of the gap closed by rung h, the convergence to a shrunken value and the half-life (Equation (7)).
IC_t, $\overline{\mathrm{IC}}$	The daily Spearman information coefficient and its mean over sessions (Equation (2)).
R^2_{OOS}, ΔR^2	The out-of-sample R^2 and a step's increment (Equation (8)).

Table A2, continued

Symbol	Meaning
Acc_m, p_0	Company-identification accuracy and the matched baseline (Equation (9)).
ECE	Expected calibration error of the top class of three (a fall, no move or a rise, split at ± 0.5 typical moves against e_5), 15 equal-width bins.
DSR, SR_0	The deflated-Sharpe probability and its benchmark (Equation (A1)).
H1, H1b, H2, H3, H4	Primary hypotheses (Holm at 0.05).
H5, H6, H7, H2-C	Secondary hypotheses (Benjamini–Hochberg at $q = 0.10$): explaining moves, style, the instruments, convergence.
F1 to F7	The forward test's hypotheses (Section 11).

C The deflated Sharpe ratio

For a book with T daily net returns, a per-period Sharpe ratio $\widehat{\mathrm{SR}}$, skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$, the deflated-Sharpe probability (Bailey and López de Prado, 2014) is computed as

$$\mathrm{DSR} = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{\mathrm{SR}}^2}}\right), \qquad \mathrm{SR}_0 = \sqrt{\mathcal{V}}\left[(1-\gamma)\,\Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma\,\Phi^{-1}\left(1 - \frac{1}{N\mathrm{e}}\right)\right], \tag{A1}$$

where Φ is the standard normal distribution function, γ the Euler–Mascheroni constant, N the number of trials and $\mathcal{V} = (1 + \widehat{\mathrm{SR}}^2/2)/(T-1)$ the variance of a Sharpe-ratio estimate (Lo, 2002); the analytic variance is used because most of the ledger's other trials are information coefficients, recalls and losses, not Sharpe ratios. SR_0 is the Sharpe ratio that the best of N independent trials without skill would reach in expectation, so it rises with N: a book must beat it, not zero. With the 545 trials of the ledger when the pre-registered strategies were evaluated and 139 test days, SR_0 is 0.26 per day for the convergence strategy, far above any book's daily Sharpe ratio. The exploratory books of Section 10 were deflated with all 1,620 trials of the ledger after the exploratory study, which raises the bar further.

D Additional pre-registered results

This appendix gives the pre-registered results that the main text cites but does not show: every model's IC and cost (Table A3), the robustness variants (Figure A1), the heterogeneity cuts (Table A4), the secondary hypotheses (Table A5) and calibration (Table A6 and Figure A2). Further tables and figures are in the replication package.

Table A3: Model comparison: every val-B IC with z_{cc} that is positive falls on the test (Laya-Fin 0.065 to 0.020); the cost and the latency.

Model	IC val-B	t	IC test	t	Events	USD/1,000	p50 (ms)
Laya-Fin	0.065	3.64	0.020	1.67	13,748	0.000	—
anchor	−0.020	−1.52	0.008	0.83	13,748	0.000	—
Laya zero-shot	0.029	1.94	0.005	0.37	13,748	0.000	536
d1-ultra	0.097	5.38	0.039	3.23	13,748	0.063	288
span-02[‡]	0.099	5.83	0.041	3.10	13,748	0.028	2,262
drex-v2.0	0.075	4.10	0.044	3.70	13,748	0.045	680
vendor sentiment[†]	0.087	4.39	0.047	3.36	13,748	—	—
bge-ridge	0.002	0.10	0.004	0.45	13,748	—	—
tags-GBM[†]	0.108	6.03	0.055	3.91	13,748	—	—
LLM baseline	0.052	1.93	0.001	0.02	1,977	—	—
price-only	0.004	0.33	−0.003	−0.26	13,748	—	—
Laya-Fin, LLM's 2,000-event subsample	—	—	−0.012	−0.37	1,977	—	—
LLM baseline, its subsample	—	—	0.001	0.02	1,977	—	—
price-only (D2, IC with y_2)	—	—	0.012	1.18	17,188	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
[†]Vendor-label baseline (vendor sentiment, tags-GBM): the labels were produced retrospectively, so its IC is a possibly look-ahead-contaminated upper bound, not the benchmark; the clean benchmarks are bge-ridge, price-only, the LLM baseline and the random-signal placebo.
[‡]span-02: after the freeze, a stability check that repeats earlier requests (the canary) fired for it; its build was unchanged, and production went ahead under a rule recorded in the deviations log.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the pre-registered contamination screens, so their results are model-specific.

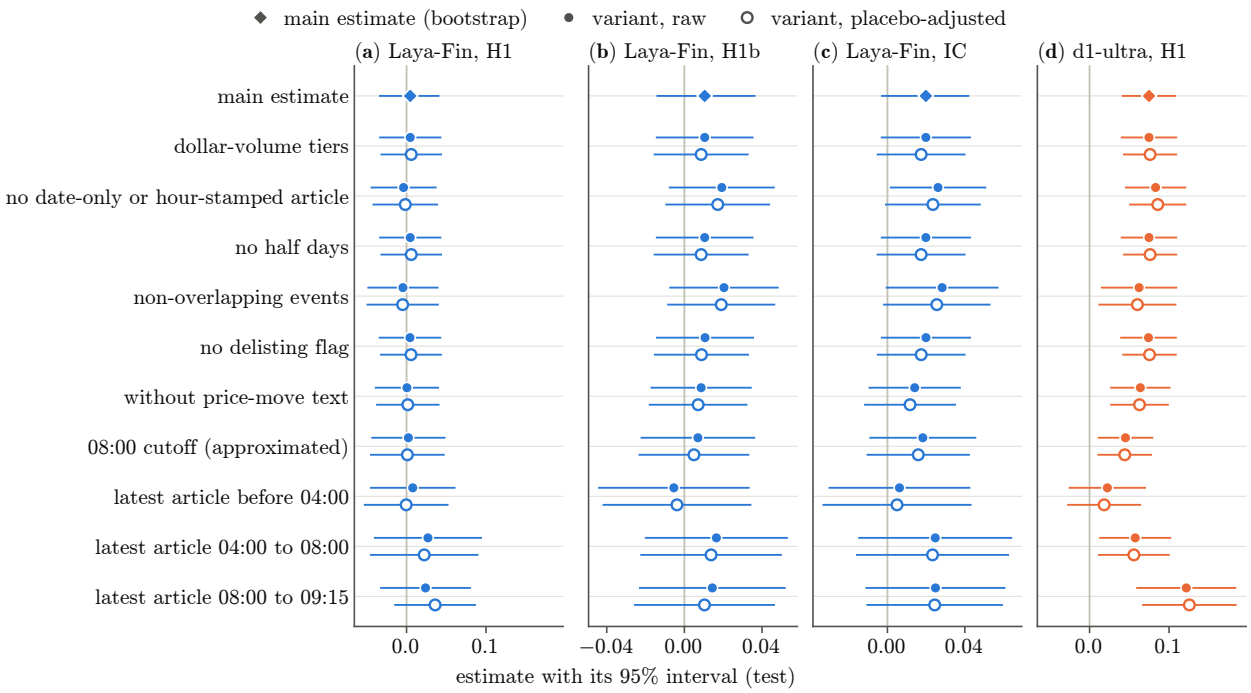


Figure A1: No pre-registered robustness variant changes a conclusion: Laya-Fin's H1 and H1b stay within their intervals of zero, and d1-ultra's H1 stays positive in every variant, from 0.022 when the latest article precedes 04:00 to 0.122 when it arrives between 08:00 and 09:15. Test period, D0 events. Each variant raw (filled) and with placebo-adjusted targets (hollow), below the main estimate (diamond, from the stationary block bootstrap), with 95% intervals (±1.96 Newey–West standard errors for the variants). (a) and (d), both H1, share a scale; (b) and (c) have their own. d1-ultra is model-specific.

Table A4: Heterogeneity of H2: no cut differs (Laya-Fin's Wald p from 0.16 to 0.90).

Model, cut	Events	Wald F	p	BH q	Δb	SE	Δc	SE
Laya-Fin, Top 100	4,516	1.83	0.164	0.741	0.490	(0.281)	−0.025	(0.021)
Laya-Fin, 101 to 250	5,121	0.10	0.904	0.904	−0.008	(0.258)	−0.007	(0.016)
Laya-Fin, 251 to 500	7,551	1.71	0.185	0.741	−0.326	(0.253)	0.023	(0.016)
Laya-Fin, quantified	15,030	0.66	0.521	0.833	0.470	(0.609)	0.022	(0.026)
Laya-Fin, small move	9,795	0.29	0.751	0.859	−0.193	(0.255)	0.005	(0.035)
Laya-Fin, press wire	9,655	0.75	0.477	0.833	−0.058	(0.330)	−0.024	(0.020)
Laya-Fin, low usual coverage	8,387	0.44	0.645	0.859	0.040	(0.247)	0.014	(0.016)
Laya-Fin, busy day	7,805	0.93	0.398	0.833	−0.194	(0.263)	−0.019	(0.019)
anchor, Top 100	4,516	2.18	0.117	0.469	6.973	(4.120)	−0.023	(0.020)
anchor, 101 to 250	5,121	0.52	0.594	0.679	−2.559	(2.719)	−0.007	(0.016)
anchor, 251 to 500	7,551	1.18	0.312	0.679	−2.500	(3.035)	0.021	(0.015)
anchor, quantified	15,030	2.84	0.062	0.469	7.109	(3.308)	0.024	(0.026)
anchor, small move	9,795	0.01	0.992	0.992	0.018	(2.650)	0.004	(0.035)
anchor, press wire	9,655	0.79	0.457	0.679	−1.650	(3.084)	−0.024	(0.020)
anchor, low usual coverage	8,387	0.75	0.477	0.679	2.272	(2.652)	0.014	(0.016)
anchor, busy day	7,805	0.58	0.561	0.679	1.140	(2.570)	−0.020	(0.019)
d1-ultra, Top 100	4,516	1.10	0.337	0.719	0.059	(0.060)	−0.024	(0.020)
d1-ultra, 101 to 250	5,121	0.88	0.418	0.719	0.063	(0.055)	−0.010	(0.016)
d1-ultra, 251 to 500	7,551	3.36	0.038	0.301	−0.095	(0.050)	0.026	(0.015)
d1-ultra, quantified	15,030	0.53	0.591	0.719	0.014	(0.062)	0.023	(0.027)
d1-ultra, small move	9,795	0.15	0.857	0.857	0.023	(0.044)	0.002	(0.035)
d1-ultra, press wire	9,655	0.67	0.515	0.719	−0.021	(0.049)	−0.022	(0.020)
d1-ultra, low usual coverage	8,387	2.52	0.084	0.336	0.096	(0.046)	0.010	(0.016)
d1-ultra, busy day	7,805	0.47	0.629	0.719	0.002	(0.043)	−0.019	(0.019)

Wald test of $\hat{N} \cdot 1[\text{cut}]$ and $G \cdot 1[\text{cut}]$ added to H2 at D2; BH over each model's eight cuts. Events: the D2 events in the cut.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the pre-registered contamination screens, so their results are model-specific.

Table A5: Secondary hypotheses: only H5 survives Benjamini–Hochberg for Laya-Fin ($q = 0.058$).

Model, test	Estimate	95% CI	SE	p	BH q	Pooled q
Laya-Fin, H5	0.0025	—	—	0.010	0.058	0.078
Laya-Fin, H6	—	—	—	—	—	—
Laya-Fin, H7	0.0146	[−0.0068, 0.0363]	—	0.097	0.238	0.317
Laya-Fin, H2-C, κ at h^*	0.2588	—	(0.2299)	0.131	0.238	0.317
Laya-Fin, H2-C, κ increase	0.0182	[−0.3614, 0.4903]	—	0.467	0.561	0.748
Laya-Fin, H2-C, λ_{t+5}	0.0265	—	(0.0264)	0.159	0.238	0.317
Laya-Fin, H2-C, naive tracking	1.0007	[0.9995, 1.0018]	—	0.867	0.867	0.946
Laya-Fin, A3.4	—	—	—	—	—	—
anchor, H5	0.0003	—	—	0.072	0.358	—
anchor, H6	—	—	—	—	—	—
anchor, H7	0.0024	[−0.0294, 0.0355]	—	0.442	—	—
anchor, H2-C, κ at h^*	0.0798	—	(0.1574)	0.307	0.511	—
anchor, H2-C, κ increase	−0.0031	[−0.2669, 0.4215]	—	0.514	0.643	—
anchor, H2-C, λ_{t+5}	0.0270	—	(0.0266)	0.156	0.390	—
anchor, H2-C, naive tracking	1.0020	[1.0002, 1.0036]	—	0.992	0.992	—
anchor, A3.4	—	—	—	—	—	—
d1-ultra, H5	0.0082	—	—	<0.001	<0.001	—
d1-ultra, H6	0.0104	—	(0.0192)	0.705	0.946	0.940
d1-ultra, H7	—	—	—	—	—	—
d1-ultra, H2-C, κ at h^*	0.3645	—	(0.2152)	0.046	0.163	—
d1-ultra, H2-C, κ increase	−0.1805	[−0.5343, 0.2158]	—	0.834	0.946	—
d1-ultra, H2-C, λ_{t+5}	0.0216	—	(0.0262)	0.206	0.480	—
d1-ultra, H2-C, naive tracking	1.0005	[0.9993, 1.0017]	—	0.781	0.946	—
d1-ultra, A3.4	−0.0619	—	(0.0382)	0.946	0.946	0.946

—: the source table has no value for the row; the statistic does not apply to it or was not computed.

BH at $q = 0.10$ within each instrument; the pooled family adds d1-ultra's H6 and A3.4 to Laya-Fin's. The anchor's H7 is reported outside its family, so it has no q. Each test has either an SE or a 95% CI, as the frozen analysis reports it.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the pre-registered contamination screens, so their results are model-specific.

Table A6: Calibration of the three-class forecasts: the trained models are calibrated but barely informative (Laya-Fin's ECE 0.057), the API models overconfident (ECE 0.209 to 0.361).

Model, target	ECE	Log-loss skill	Brier skill	AUC	T	ECE before	ECE after
Laya-Fin, e_5	0.057	0.001	0.000	—	—	—	—
Laya-Fin, z_{cc}	0.033	0.010	0.010	—	—	—	—
Laya-Fin, $\|z_{cc}\| \geq 1$	—	—	—	0.60	—	—	—
Laya-Fin, e_5, temperature scaling	—	—	—	—	1.00	0.058	0.057
anchor, e_5	0.048	0.002	0.001	—	—	—	—
anchor, z_{cc}	0.016	0.015	0.015	—	—	—	—
anchor, $\|z_{cc}\| \geq 1$	—	—	—	0.58	—	—	—
anchor, e_5, temperature scaling	—	—	—	—	0.97	0.011	0.048
Laya zero-shot, e_5	0.176	−0.156	−0.110	—	—	—	—
Laya zero-shot, z_{cc}	0.179	−0.177	−0.120	—	—	—	—
Laya zero-shot, $\|z_{cc}\| \geq 1$	—	—	—	0.52	—	—	—
Laya zero-shot, e_5, temperature scaling	—	—	—	—	6.33	0.176	0.110
d1-ultra, e_5	0.260	−0.194	−0.194	—	—	—	—
d1-ultra, z_{cc}	0.249	−0.166	−0.165	—	—	—	—
d1-ultra, $\|z_{cc}\| \geq 1$	—	—	—	0.58	—	—	—
d1-ultra, e_5, temperature scaling	—	—	—	—	1.95	0.260	0.155
span-02[‡], e_5	0.361	−0.460	−0.361	—	—	—	—
span-02[‡], z_{cc}	0.342	−0.410	−0.315	—	—	—	—
span-02[‡], $\|z_{cc}\| \geq 1$	—	—	—	0.56	—	—	—
drex-v2.0, e_5	0.209	−0.135	−0.138	—	—	—	—
drex-v2.0, z_{cc}	0.204	−0.119	−0.120	—	—	—	—
drex-v2.0, $\|z_{cc}\| \geq 1$	—	—	—	0.57	—	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
The pre-open reading's three classes (a fall, no move or a rise, split at ±0.5 typical moves) against the targets; skill against val-A climatology; T the temperature fit on val-A.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
[‡]span-02: after the freeze, a stability check that repeats earlier requests (the canary) fired for it; its build was unchanged, and production went ahead under a rule recorded in the deviations log.

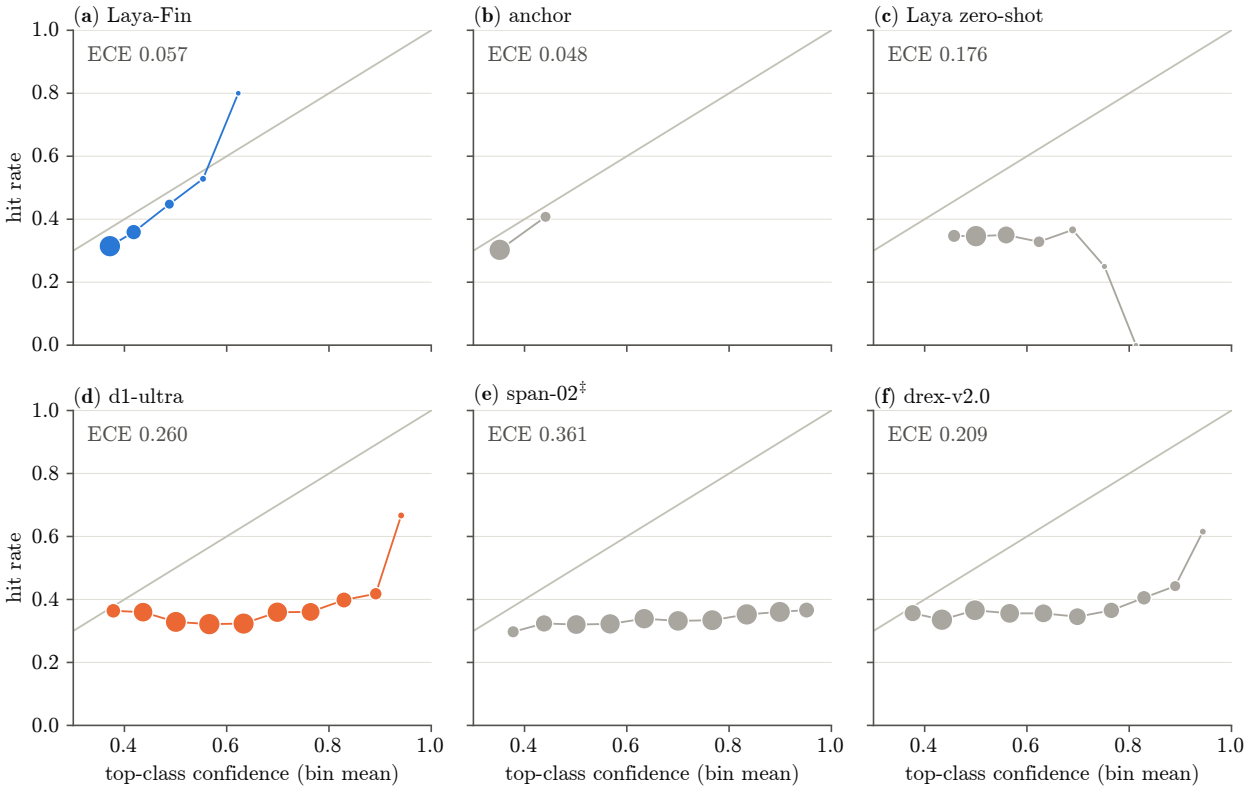


Figure A2: Laya-Fin's three-class forecasts are calibrated but compressed (expected calibration error 0.057); the commercial models are overconfident (0.209 to 0.361). Reliability of the pre-open reading's three classes (a fall, no move or a rise, split at ±0.5 typical moves) against the five-day move e_5 on the test period: the hit rate of the top class by bin of its stated probability (15 bins; the dot area grows with the number of events in the bin; the diagonal is perfect calibration). The anchor's near-constant readings sit in two bins. ‡span-02: see Figure 1. The commercial models are model-specific.

E Exploratory: the conditioning study

Figure A3 shows every (model, condition) pair of the post hoc conditioning study of Section 10, with the 8 pairs that the validation rule selected shaded. It is exploratory: the study was requested after unblinding, and the figure supports no claim.

F The replication package

The replication package is the project's repository **[repository URL to be provided by the authors]**. It holds the pre-registration with its amendments and its deviations log; the freeze record (the hash, its time and the version-control tag); the code of every stage, from the price panel and the masked news states to the model calls, the analysis and the strategies, with its tests; the committed outputs and a report of every stage; the append-only, hash-chained ledgers of API spend, trials, test-data access and data requests; and the sources of this paper. Further tables and figures are in the replication package. One command rebuilds every table, figure and caption of the paper from the committed outputs (plus three stored strategy files that the strategy stage regenerates) and checks each against a committed digest; another writes the macros through which every result number in this paper enters the text, each a cell of those tables, and checks that no result number in the sources is typed by hand. Neither needs raw data, an API, a GPU or the lock. The news text is licensed and is not redistributed; **[data availability statement to be provided by the authors]**.

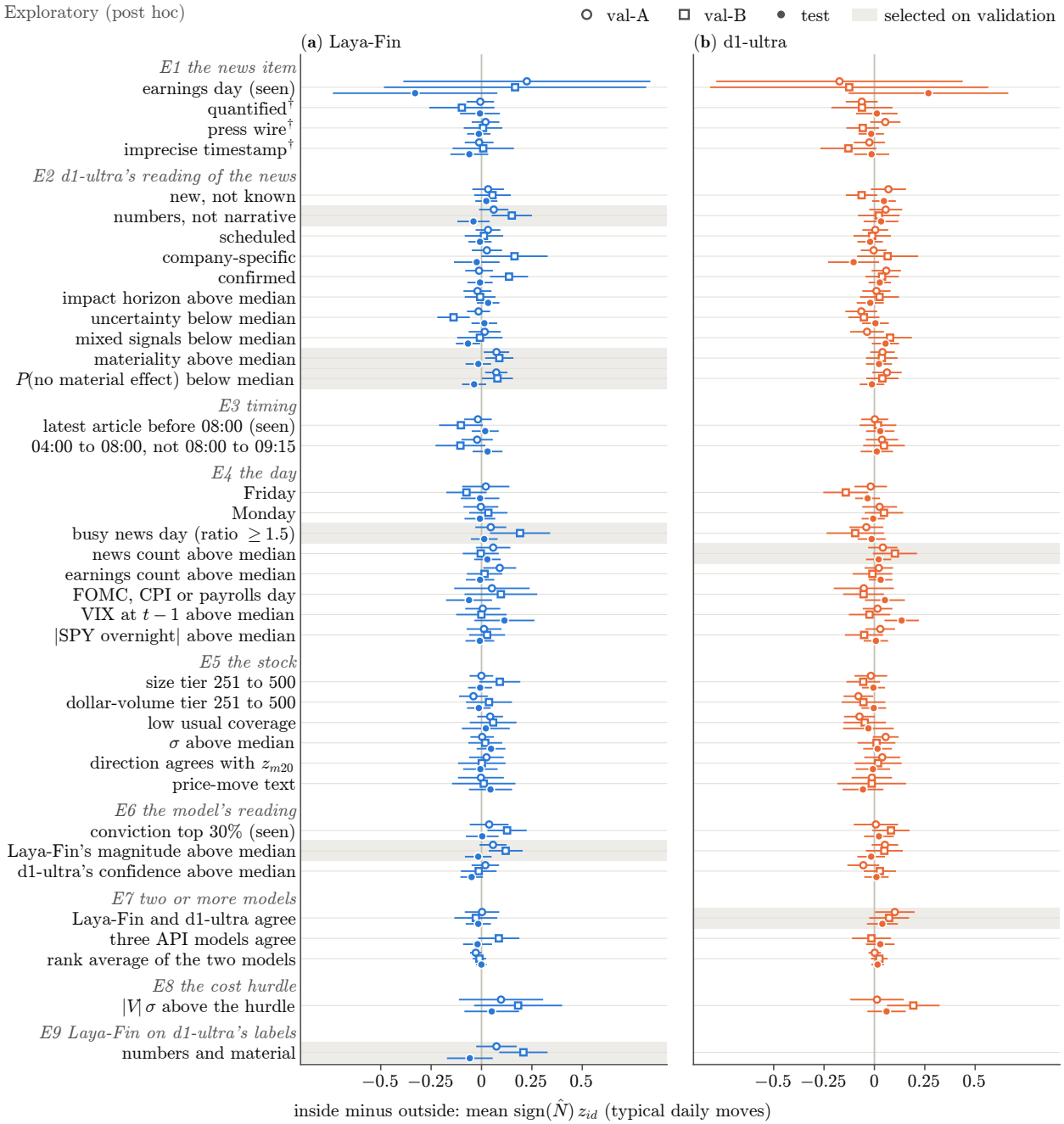


Figure A3: Exploratory (post hoc): none of the 8 conditions selected on two validation quarters held on the test; 5 changed sign. For each (model, condition) pair, the difference inside minus outside the condition of the mean sign$(\hat{N}) \cdot z_{id}$ (typical daily moves) on val-A (circles), val-B (squares) and the test (dots), with 95% intervals; shaded rows were selected by the rule fixed before the test run (positive with $t \geq 1$ in both validation quarters and a pooled $t \geq 2$). †Vendor fields are retrospective labels and only split events descriptively. "(seen)" marks conditions whose test values the authors' assistant had seen before the study, which were never selected. Requested after unblinding; no claim.